

GLP-1 Author Journey Extraction Pipeline

```
In [ ]: %pip install --upgrade pip
        %pip install pandas requests torch transformers accelerate tiktoken openai
```

Requirement already satisfied: pip in /opt/homebrew/lib/python3.11/site-packages (26.1.2)

Note: you may need to restart the kernel to use updated packages.

Requirement already satisfied: pandas in /opt/homebrew/lib/python3.11/site-packages (3.0.3)

Requirement already satisfied: requests in /opt/homebrew/lib/python3.11/site-packages (2.34.2)

Requirement already satisfied: torch in /opt/homebrew/lib/python3.11/site-packages (2.13.0)

Requirement already satisfied: transformers in /opt/homebrew/lib/python3.11/site-packages (5.14.1)

Requirement already satisfied: accelerate in /opt/homebrew/lib/python3.11/site-packages (1.14.0)

Requirement already satisfied: tiktoken in /opt/homebrew/lib/python3.11/site-packages (0.13.0)

Requirement already satisfied: openai in /opt/homebrew/lib/python3.11/site-packages (2.41.0)

Requirement already satisfied: rouge-score in /opt/homebrew/lib/python3.11/site-packages (0.1.2)

Requirement already satisfied: bert-score in /opt/homebrew/lib/python3.11/site-packages (0.3.13)

Requirement already satisfied: matplotlib in /opt/homebrew/lib/python3.11/site-packages (3.11.0)

Requirement already satisfied: numpy>=1.26.0 in /opt/homebrew/lib/python3.11/site-packages (from pandas) (2.4.6)

Requirement already satisfied: python-dateutil>=2.8.2 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: charset-normalizer<4,>=2 in /opt/homebrew/lib/python3.11/site-packages (from requests) (3.4.7)

Requirement already satisfied: idna<4,>=2.5 in /opt/homebrew/lib/python3.11/site-packages (from requests) (3.18)

Requirement already satisfied: urllib3<3,>=1.26 in /opt/homebrew/lib/python3.11/site-packages (from requests) (2.7.0)

Requirement already satisfied: certifi>=2023.5.7 in /opt/homebrew/lib/python3.11/site-packages (from requests) (2026.5.20)

Requirement already satisfied: filelock in /opt/homebrew/lib/python3.11/site-packages (from torch) (3.29.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from torch) (4.15.0)

Requirement already satisfied: setuptools>=77.0.3 in /opt/homebrew/lib/python3.11/site-packages (from torch) (81.0.0)

Requirement already satisfied: sympy>=1.13.3 in /opt/homebrew/lib/python3.11/site-packages (from torch) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /opt/homebrew/lib/python3.11/site-packages (from torch) (3.6.1)

Requirement already satisfied: jinja2 in /opt/homebrew/lib/python3.11/site-packages (from torch) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /opt/homebrew/lib/python3.11/site-packages (from torch) (2026.4.0)

Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (1.18.0)

Requirement already satisfied: packaging>=20.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from transformers) (26.2)

Requirement already satisfied: pyyaml>=5.1 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (6.0.3)

Requirement already satisfied: regex>=2025.10.22 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (2026.5.9)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (0.22.2)

Requirement already satisfied: typer in /opt/homebrew/lib/python3.11/site-packages (from transformers) (0.25.1)

Requirement already satisfied: safetensors>=0.8.0 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (0.8.0)

Requirement already satisfied: tqdm>=4.60 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (4.68.1)

Requirement already satisfied: click>=8.4.0 in /opt/homebrew/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (8.4.1)

Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /opt/homebrew/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (1.5.0)

Requirement already satisfied: httpx<1,>=0.23.0 in /opt/homebrew/lib/python3.11/site-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0.28.1)

Requirement already satisfied: anyio in /opt/homebrew/lib/python3.11/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (4.13.0)

Requirement already satisfied: httpcore==1.* in /opt/homebrew/lib/python3.11/site-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (1.0.9)

Requirement already satisfied: h11>=0.16 in /opt/homebrew/lib/python3.11/site-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transformers) (0.16.0)

Requirement already satisfied: shellingham>=1.3.0 in /opt/homebrew/lib/python3.11/site-packages (from typer->transformers) (1.5.4)

Requirement already satisfied: rich>=13.8.0 in /opt/homebrew/lib/python3.11/site-packages (from typer->transformers) (15.0.0)

Requirement already satisfied: annotated-doc>=0.0.2 in /opt/homebrew/lib/python3.11/site-packages (from typer->transformers) (0.0.4)

Requirement already satisfied: psutil in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from accelerate) (7.2.2)

Requirement already satisfied: distro<2,>=1.7.0 in /opt/homebrew/lib/python3.11/site-packages (from openai) (1.9.0)

Requirement already satisfied: jiter<1,>=0.10.0 in /opt/homebrew/lib/python3.11/site-packages (from openai) (0.15.0)

Requirement already satisfied: pydantic<3,>=1.9.0 in /opt/homebrew/lib/python3.11/site-packages (from openai) (2.13.4)

Requirement already satisfied: sniffio in /opt/homebrew/lib/python3.11/site-packages (from openai) (1.3.1)

Requirement already satisfied: annotated-types>=0.6.0 in /opt/homebrew/lib/python3.11/site-packages (from pydantic<3,>=1.9.0->openai) (0.7.0)

Requirement already satisfied: pydantic-core==2.46.4 in /opt/homebrew/lib/python3.11/site-packages (from pydantic<3,>=1.9.0->openai) (2.46.4)

Requirement already satisfied: typing-inspection>=0.4.2 in /opt/homebrew/lib/python3.11/site-packages (from pydantic<3,>=1.9.0->openai) (0.4.2)

Requirement already satisfied: absl-py in /opt/homebrew/lib/python3.11/site-packages (from rouge-score) (2.5.0)

Requirement already satisfied: nltk in /opt/homebrew/lib/python3.11/site-packages (from rouge-score) (3.10.0)

Requirement already satisfied: six>=1.14.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from rouge-score) (1.17.0)

Requirement already satisfied: contourpy>=1.0.1 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (4.63.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: pillow>=9 in /opt/homebrew/lib/python3.11/s

```

ite-packages (from matplotlib) (12.2.0)
Requirement already satisfied: pyparsing>=3 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /opt/homebrew/lib/python3.11/site-packages (from rich>=13.8.0->typer->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from rich>=13.8.0->typer->transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /opt/homebrew/lib/python3.11/site-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/homebrew/lib/python3.11/site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/homebrew/lib/python3.11/site-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: defusedxml in /opt/homebrew/lib/python3.11/site-packages (from nltk->rouge-score) (0.7.1)
Requirement already satisfied: joblib in /opt/homebrew/lib/python3.11/site-packages (from nltk->rouge-score) (1.5.3)
Note: you may need to restart the kernel to use updated packages.

```

```

In [ ]: #####
# Imports #
#####

# Standard library
import os
import re
import sys
import glob
import json
import time
import getpass
import hashlib
from pathlib import Path
from collections import defaultdict

# Data / plotting
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
from matplotlib.gridspec import GridSpec
from matplotlib.colors import TwoSlopeNorm
from matplotlib.cm import ScalarMappable
from matplotlib.lines import Line2D
from scipy.stats import fisher_exact, mannwhitneyu
from statsmodels.stats.multitest import multipletests

# Modelling / tokenisation (kept from the original environment)
import requests
import torch
import accelerate
import transformers
from transformers import pipeline
import tiktoken

# OpenAI client
from openai import OpenAI

# Evaluation metrics

```

```
import rouge_score
import bert_score
from rouge_score import rouge_scorer
from bert_score import score as bert_score
```

/opt/homebrew/lib/python3.11/site-packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
 from .autonotebook import tqdm as notebook_tqdm

```
In [ ]: #####
# Defining paths and constants #
#####

base_dir = "/Users/telissatitley/llm_env/GLP1_Thesis"
subreddits_root = f"{base_dir}/Subreddits"

subsample_dir = f"{subreddits_root}/Subsamples"
test_output_dir = f"{subreddits_root}/Test"
gpt_result_dir = f"{subreddits_root}/GPT_Results"

for d in [subsample_dir, test_output_dir, gpt_result_dir]:
    os.makedirs(d, exist_ok=True)

# Manually annotated files
few_shot_annotated_path = f"{test_output_dir}/few_shot_author_annotated_e
gold_standard_annotated_path = f"{test_output_dir}/gold_standard_evaluati

# API key request line
if not os.environ.get("OPENAI_API_KEY"):
    os.environ["OPENAI_API_KEY"] = getpass.getpass("Enter OpenAI API key:
```

1. Load raw subreddit json files

Semaglutide

```
In [6]: sema_oz_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subred
sema_oz_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subred
sema_weg_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subre
sema_weg_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subre
sema_gen_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subre
sema_gen_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subre
```

Retatrutide

```
In [7]: reta_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddit
reta_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddit
```

2. Clean and standardise

```
In [8]: #Check all posts and comments alignment across each indication.
```

```

keep_cols = ['author', 'archived', 'body', 'created_utc', 'id'] #defining

def align_posts(df):
    df = df.rename(columns={'selftext': 'body'}) #updating posts 'selftex
    return df[keep_cols]

def align_comments(df):
    return df[keep_cols]

#Aligning relevant tags from posts and comments section for each drug ind
# Semaglutide
# Ozempic
sema_oz_p = align_posts(sema_oz_p)
sema_oz_c = align_comments(sema_oz_c)

# Wegovy
sema_weg_p = align_posts(sema_weg_p)
sema_weg_c = align_comments(sema_weg_c)

# Generic
sema_gen_p = align_posts(sema_gen_p)
sema_gen_c = align_comments(sema_gen_c)

# Retatrutide
reta_p = align_posts(reta_p)
reta_c = align_comments(reta_c)

#Sense checking only relevant fields remain.
# Retatrutide
print(sema_oz_p.columns.tolist())
print(sema_oz_c.columns.tolist())

```

```

['author', 'archived', 'body', 'created_utc', 'id']
['author', 'archived', 'body', 'created_utc', 'id']

```

```

In [9]: sema_oz_p["source"] = "post"
        sema_oz_c["source"] = "comment"

        sema_weg_p["source"] = "post"
        sema_weg_c["source"] = "comment"

        sema_gen_p["source"] = "post"
        sema_gen_c["source"] = "comment"

        reta_p["source"] = "post"
        reta_c["source"] = "comment"

```

```

In [10]: #Removing automoderator, deleted and removed authors.

def clean_data(df):
    df['body'] = df['body'].astype(str)

    return df[
        (df['author'].notna()) &
        (~df['author'].isin(['[deleted]', '[removed]', 'AutoModerator']))
        (~df['body'].isin(['[removed]', '[deleted]'])) &
        (df['body'].str.strip() != '')
    ]

# Ozempic

```

```

sema_oz_c = clean_data(sema_oz_c)
sema_oz_p = clean_data(sema_oz_p)

# Wegovy
sema_weg_c = clean_data(sema_weg_c)
sema_weg_p = clean_data(sema_weg_p)

# Generic Semaglutide
sema_gen_c = clean_data(sema_gen_c)
sema_gen_p = clean_data(sema_gen_p)

# Retatrutide
reta_c = clean_data(reta_c)
reta_p = clean_data(reta_p)

```

In [11]: *## Merging posts and comments for ozempic, wegovy, generic semaglutide an*

```

# Ozempic
sema_oz = pd.concat([sema_oz_p, sema_oz_c], ignore_index=True)
# Wegovy
sema_weg = pd.concat([sema_weg_p, sema_weg_c], ignore_index=True)
# Generic semaglutide
sema_gen = pd.concat([sema_gen_p, sema_gen_c], ignore_index=True)

# Retatrutide
reta = pd.concat([reta_p, reta_c], ignore_index=True)

sema_oz["drug"] = "ozempic"
sema_weg["drug"] = "wegovy"
sema_gen["drug"] = "semaglutide"
reta["drug"] = "retatrutide"

```

In [12]: *#Checking for archived users.*

```

print(sema_oz['archived'].value_counts(dropna=False))
print(sema_weg['archived'].value_counts(dropna=False))
print(sema_gen['archived'].value_counts(dropna=False))
print(reta['archived'].value_counts(dropna=False))

```

```

archived
False    943779
True       1706
Name: count, dtype: int64
archived
False    34697
Name: count, dtype: int64
archived
False    973128
Name: count, dtype: int64
archived
False    803706
Name: count, dtype: int64

```

In [13]: *#Sense checking for any empty posts/comments in body*

```

sema_oz['body'].isna().sum()
#Output: np.int64(0)
sema_weg['body'].isna().sum()
#Output: np.int64(0)
sema_gen['body'].isna().sum()

```

```

#Output: np.int64(0)
reta['body'].isna().sum()
#Output: np.int64(0)

#Removing duplicate bodies of texts
sema_oz = sema_oz.drop_duplicates(subset=["body"])
sema_weg = sema_weg.drop_duplicates(subset=["body"])
sema_gen = sema_gen.drop_duplicates(subset=["body"])
reta = reta.drop_duplicates(subset=["body"])

print("oz:", len(sema_oz))
print("weg:", len(sema_weg))
print("gen:", len(sema_gen))
print("reta:", len(reta))

```

```

oz: 898471
weg: 34174
gen: 923001
reta: 764833

```

```

In [14]: # Ozempic
print("Ozempic:", sema_oz.shape)
#Output: Ozempic: (898471, 6)

# Wegovy
print("Wegovy:", sema_weg.shape)
#Output: Wegovy: (34174, 6)

# Generic Semaglutide
print("Generic Semaglutide:", sema_gen.shape)
#Output: Generic Semaglutide: (923001, 6)

# Retatrutide
print("Retatrutide:", reta.shape)
#Output: Retatrutide: (764833, 6)

```

```

Ozempic: (898471, 7)
Wegovy: (34174, 7)
Generic Semaglutide: (923001, 7)
Retatrutide: (764833, 7)

```

```

In [ ]: sema_oz.head()

```

```

In [16]: #Handling NaNs to prevent dropping relevant columns.
sema_oz = sema_oz[sema_oz['archived'].fillna(False) == False]
sema_weg = sema_weg[sema_weg['archived'].fillna(False) == False]
sema_gen = sema_gen[sema_gen['archived'].fillna(False) == False]
reta = reta[reta['archived'].fillna(False) == False]

```

```

In [17]: print("oz:", len(sema_oz))
print("weg:", len(sema_weg))
print("gen:", len(sema_gen))
print("reta:", len(reta))

```

```

oz: 896774
weg: 34174
gen: 923001
reta: 764833

```



```
In [ ]: #Dropping archived column completely.
```

```
sema_oz = sema_oz.drop(columns=['archived'])
sema_weg = sema_weg.drop(columns=['archived'])
sema_gen = sema_gen.drop(columns=['archived'])
reta = reta.drop(columns=['archived'])
```

```
#Sense checking each indication.
```

```
print(sema_oz.columns)
print(sema_weg.columns)
print(sema_gen.columns)
print(reta.columns)
```

```
#Output: Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'],
```

```
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

```
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

```
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

```
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

Convert UTC timestamps to datetime

```
In [19]: #Convert UTC to standard date format
```

```
sema_oz['created_utc'] = pd.to_datetime(sema_oz['created_utc'], unit='s')
sema_weg['created_utc'] = pd.to_datetime(sema_weg['created_utc'], unit='s')
sema_gen['created_utc'] = pd.to_datetime(sema_gen['created_utc'], unit='s')
reta['created_utc'] = pd.to_datetime(reta['created_utc'], unit='s')
```

```
#Sense-checking change complete
```

```
sema_oz[['created_utc']].head()
```

```
Out [19]:
```

	created_utc
0	2019-12-04 20:01:54
1	2021-05-09 03:51:59
2	2021-05-09 13:41:26
3	2021-05-11 22:29:54
4	2021-05-12 12:49:08

3. Filtering for longitudinal unique authors

```
In [ ]: output_dir = "/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Subsamp
os.makedirs(output_dir, exist_ok=True)
```

```
start_date = pd.Timestamp("2019-12-04")
end_date = pd.Timestamp("2026-06-10 22:50:09")
```

```

def make_longitudinal_subsample(df, name):

    # 1. Filter date range
    sub_df = df.loc[df["created_utc"].between(start_date, end_date)].copy()

    total_rows = len(sub_df)
    total_authors = sub_df["author"].nunique()

    # 2. Keep only repeat authors
    repeat_authors = (sub_df["author"].value_counts().loc[lambda x: x > 1])

    sub_df = sub_df[sub_df["author"].isin(repeat_authors)].copy()

    repeat_author_count = len(repeat_authors)

    # 3. Sort chronologically within author
    sub_df = sub_df.sort_values(["author", "created_utc"])
    grouped = sub_df.groupby("author")

    # 4. Add longitudinal metadata
    sub_df["n_author_posts"] = (sub_df.groupby("author")["author"].transform("count"))
    sub_df["journey_step"] = (sub_df.groupby("author").cumcount() + 1)
    sub_df["first_post_date"] = grouped["created_utc"].transform("min")
    sub_df["days_since_first_post"] = (sub_df["created_utc"] - sub_df["first_post_date"])

    retained_rows = len(sub_df)

    # Save
    base_path = f"{output_dir}/sub_{name}_longitudinal"

    sub_df.to_json(f"{base_path}.jsonl", orient="records", lines=True, date_format="iso")
    sub_df.to_csv(f"{base_path}.csv", index=False)

    # Summary
    print(f"\n{name.upper()}")
    print(f"Total rows: {total_rows:,}")
    print(f"Unique authors: {total_authors:,}")
    print(f"Repeat authors: {repeat_author_count:,}")
    print(f"Retained rows: {retained_rows:,}")

    return sub_df

```

```

In [21]: sub_sem_oz = make_longitudinal_subsample(sema_oz, "oz")
sub_sem_weg = make_longitudinal_subsample(sema_weg, "weg")
sub_sem_gen = make_longitudinal_subsample(sema_gen, "gen")
sub_reta = make_longitudinal_subsample(reta, "reta")

```

OZ

Total rows: 896,774
Unique authors: 98,544
Repeat authors: 58,498
Retained rows: 856,728

WEG

Total rows: 34,174
Unique authors: 8,277
Repeat authors: 4,479
Retained rows: 30,376

GEN

Total rows: 923,001
Unique authors: 114,308
Repeat authors: 67,054
Retained rows: 875,747

RETA

Total rows: 764,833
Unique authors: 67,483
Repeat authors: 41,643
Retained rows: 738,993

In [22]: *#Saving for reproducibility*

```
sema_oz.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Sema  
sema_weg.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Sem  
sema_gen.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Sem  
reta.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatru
```

```
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/331779321  
2.py:3: Pandas4Warning: The default 'epoch' date format is deprecated and  
will be removed in a future version, please use 'iso' date format instead.  
sema_oz.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Sem  
aglutide/Clean/sema_oz.jsonl',orient='records', lines=True)  
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/331779321  
2.py:4: Pandas4Warning: The default 'epoch' date format is deprecated and  
will be removed in a future version, please use 'iso' date format instead.  
sema_weg.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Se  
maglutide/Clean/sema_weg.jsonl',orient='records', lines=True)  
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/331779321  
2.py:5: Pandas4Warning: The default 'epoch' date format is deprecated and  
will be removed in a future version, please use 'iso' date format instead.  
sema_gen.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Se  
maglutide/Clean/sema_gen.jsonl',orient='records', lines=True)  
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/331779321  
2.py:6: Pandas4Warning: The default 'epoch' date format is deprecated and  
will be removed in a future version, please use 'iso' date format instead.  
reta.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatru  
utide/Clean/reta.jsonl',orient='records', lines=True)
```

4. Keywords and terms search: lexical screening

```
In [ ]: seed_terms = {"journey": ["week", "month", "day", "days", "year", "starte  
weight": ["lost", "down", "gained", "lbs", "kg", "weight", "increased",
```

```

"dose": ["dose", "dosage", "mg", "0.25", "0.5", "1mg", "2.4mg", "upped do
"experience": ["side effect", "nausea", "appetite", "food noise", "binge"
"height": ["ft", "cm", "inches", "centimetres"],
"age": ["years old", "age", "yo", "y/o"],
"sex": ["female", "male", "woman", "man", "cis", "sex", "gender"],
"exercise": ["exercise", "workout", "gym", "training", "sedentary", "acti
}

#Removing seed search sensitivity
seed_terms_lower = {
    category: [term.lower() for term in terms]
    for category, terms in seed_terms.items()
}

def flag_terms(text, terms):
    text = str(text).lower()
    return any(term in text for term in terms)

def add_seed_flags(df, seed_terms_lower):
    df = df.copy()

    # Create category flags
    for category, terms in seed_terms_lower.items():
        df[category] = df["body"].apply(lambda x: flag_terms(x, terms))

    # Calculate relevance score
    df["relevance_score"] = df[list(seed_terms_lower.keys())].sum(axis=1)

    # Apply relevance threshold
    df["relevance"] = df["relevance_score"] >= 6

    return df

datasets = {
    "oz": sub_sem_oz,
    "weg": sub_sem_weg,
    "sema": sub_sem_gen,
    "reta": sub_reta
}

# Apply seed term flags to each dataset
for name, df in datasets.items():
    datasets[name] = add_seed_flags(df, seed_terms_lower)

# Summary output
for name, df in datasets.items():

    print(f"\n{'=' * 50}")
    print(f"{name.upper()}")
    print(f"\n{'=' * 50}")

    print("\nCounts")
    print(df[list(seed_terms_lower.keys())].sum())

    print("\nProportions")
    print(df[list(seed_terms_lower.keys())].mean())

    print(
        f"\nRelevant Posts: "

```

```
f"{df['relevance'].sum():,} / {len(df):,} "  
f"({df['relevance'].mean():.1%})"  
)
```

=====

OZ

=====

Counts

journey	372424
weight	264878
dose	148128
experience	141397
height	91815
age	426112
sex	84724
exercise	51342

dtype: int64

Proportions

journey	0.434705
weight	0.309174
dose	0.172900
experience	0.165043
height	0.107169
age	0.497371
sex	0.098893
exercise	0.059928

dtype: float64

Relevant Posts: 21,958 / 856,728 (2.6%)

=====

WEG

=====

Counts

journey	15955
weight	10281
dose	7288
experience	7390
height	4083
age	15278
sex	2874
exercise	1732

dtype: int64

Proportions

journey	0.525250
weight	0.338458
dose	0.239926
experience	0.243284
height	0.134415
age	0.502963
sex	0.094614
exercise	0.057019

dtype: float64

Relevant Posts: 1,143 / 30,376 (3.8%)

=====

SEMA

=====

```
Counts
journey      391296
weight       277282
dose         130477
experience    139072
height       93497
age          433538
sex          77949
exercise     55069
dtype: int64
```

```
Proportions
journey      0.446814
weight       0.316623
dose         0.148989
experience    0.158804
height       0.106763
age          0.495049
sex          0.089009
exercise     0.062882
dtype: float64
```

Relevant Posts: 19,957 / 875,747 (2.3%)

```
=====
META
=====
```

```
Counts
journey      265962
weight       162428
dose         172547
experience    86179
height       62870
age          350079
sex          56179
exercise     49101
dtype: int64
```

```
Proportions
journey      0.359898
weight       0.219796
dose         0.233489
experience    0.116617
height       0.085075
age          0.473724
sex          0.076021
exercise     0.066443
dtype: float64
```

Relevant Posts: 15,991 / 738,993 (2.2%)

5. Combined longitudinal pool, anonymised author IDs, and candidate authors

```
In [ ]: # Combine all longitudinal drug-group datasets
longitudinal_pool = pd.concat([df.assign(drug_group=drug_group)
```

```

        for drug_group, df in datasets.items()],
        ignore_index=True)

# Ensure the date column is datetime
longitudinal_pool["created_utc"] = pd.to_datetime(
    longitudinal_pool["created_utc"],
    errors="coerce"
)

# Sort posts chronologically within each author and drug group
longitudinal_pool = (
    longitudinal_pool
    .sort_values(
        [
            "drug_group",
            "author",
            "created_utc"
        ]
    ).reset_index(drop=True))

print(f"Combined longitudinal posts: " f"{len(longitudinal_pool):,}")

```

Combined longitudinal posts: 2,501,844

```

In [ ]: def create_author_id(drug_group, author):
        """
        Anonymising author IDs using SHA256.
        """
        author_key = f"{drug_group}|{author}"
        author_hash = hashlib.sha256(author_key.encode("utf-8")).hexdigest()
        return f"{drug_group}_{author_hash}"

private_key_path = f"{test_output_dir}/private_author_key.csv"

# Checking for old author_id exists
longitudinal_pool = longitudinal_pool.drop(columns=["author_id"], errors=

if os.path.exists(private_key_path):
    # ---- Existing key found: LOAD it, do not regenerate ----
    private_author_key = pd.read_csv(private_key_path)

    longitudinal_pool = longitudinal_pool.merge(
        private_author_key,
        on=["drug_group", "author"],
        how="left",
    )

# Only hash authors that are genuinely new (absent from the saved key
missing = longitudinal_pool["author_id"].isna()
if missing.any():
    longitudinal_pool.loc[missing, "author_id"] = (
        longitudinal_pool.loc[missing]
        .apply(lambda r: create_author_id(r["drug_group"], r["author"]
    )
    private_author_key = (
        longitudinal_pool[["drug_group", "author", "author_id"]]
        .drop_duplicates()
    )
    private_author_key.to_csv(private_key_path, index=False)

```



```

        print(f"Loaded existing key and added {int(missing.sum())} new au
    else:
        print(f"Loaded existing author key ({len(private_author_key):,} a
              "- no new ids generated")

    else:
        # ---- Generate and save the key ----
        print("No private_author_key.csv found - generating author ids for th
        longitudinal_pool["author_id"] = longitudinal_pool.apply(
            lambda r: create_author_id(r["drug_group"], r["author"]), axis=1
        )
        private_author_key = (
            longitudinal_pool[["drug_group", "author", "author_id"]]
            .drop_duplicates()
        )
        private_author_key.to_csv(private_key_path, index=False)
        print(f"Saved new author key: {private_author_key['author_id'].nunique

longitudinal_pool = longitudinal_pool.sort_values(
    ["drug_group", "author_id", "created_utc"]
).copy()

longitudinal_pool["post_number"] = (
    longitudinal_pool.groupby(["drug_group", "author_id"]).cumcount().add
)

print(f"Unique author journeys: {longitudinal_pool['author_id'].nunique()

```

Loaded existing author key (171,674 authors) – no new ids generated
 Unique author journeys: 171,674

```

In [ ]: author_summary = (
    longitudinal_pool
    .groupby(["drug_group", "author_id"], as_index=False)
    .agg(n_posts=("id", "size"),
         first_post_date=("created_utc", "min"),
         last_post_date=("created_utc", "max"),
         has_candidate_post=("relevance", "any"),
         max_relevance_score=( "relevance_score", "max")
    )

    )

author_summary["days_between_first_and_last_post"] =
(author_summary["last_post_date"] - author_summary["first_post_date"]).dt

candidate_authors =
author_summary[(author_summary["n_posts"] >= 2) &
               (author_summary["days_between_first_and_last_post"] >= 14)
               (author_summary["has_candidate_post"])]
               .copy()

print(candidate_authors.groupby("drug_group")
      .agg(candidate_authors=("author_id", "nunique"),

            total_posts=("n_posts", "sum"),
            median_posts=("n_posts", "median"),
            maximum_posts=("n_posts", "max")
            )
      )

```

	candidate_authors	total_posts	median_posts	maximum_posts
drug_group				
oz	8530	432078	18.0	8794
reta	5878	390628	24.0	6296
sema	9079	391685	17.0	5192
weg	527	9723	10.0	242

Journey text builder

```
In [ ]: def build_annotation_input(author_posts):
        """
        Convert one author journey into prompt compatible text.
        """
        author_posts = author_posts.sort_values("created_utc")

        sections = []
        for row in author_posts.itertuples():
            sections.append(
                f"""POST {row.post_number}

DATE:
{row.created_utc}

TEXT:
{row.body}
"""
            )

        return "\n".join(sections)
```

6. Load fixed few-shot and gold-standard annotation sets

```
In [28]: # Load manually annotated few-shot examples (fixed - identical on every r

assert os.path.exists(few_shot_annotated_path), (
    f"Missing annotated few-shot file: {few_shot_annotated_path}"
)

with open(few_shot_annotated_path, "r", encoding="utf-8") as f:
    few_shot_examples = json.load(f)

print(f"Loaded {len(few_shot_examples)} annotated few-shot author journey
```

Loaded 8 annotated few-shot author journeys

```
In [29]: # Load manually annotated gold-standard evaluation set (fixed - identical

assert os.path.exists(gold_standard_annotated_path), (
    f"Missing annotated gold-standard file: {gold_standard_annotated_path}"
)

with open(gold_standard_annotated_path, "r", encoding="utf-8") as f:
    gold_standard_examples = json.load(f)

print(f"Loaded {len(gold_standard_examples)} annotated gold-standard jour
```

Loaded 8 annotated gold-standard journeys

```
In [30]: # Derive the author splits directly from the annotated files
# (instead of re-sampling with random_state, which regenerated new sets w
# upstream data changed)

few_shot_author_ids = {ex["author_id"] for ex in few_shot_examples}

evaluation_authors = pd.DataFrame(
    [
        {"drug_group": ex["drug_group"], "author_id": ex["author_id"]}
        for ex in gold_standard_examples
    ]
).drop_duplicates()

print(f"Few-shot authors: {len(few_shot_author_ids)}")
print(f"Evaluation authors: {len(evaluation_authors)}")
```

Few-shot authors: 8
Evaluation authors: 8

```
In [ ]: # Remove few-shot authors from the candidate author pool
remaining_candidate_authors = candidate_authors[
    ~candidate_authors["author_id"].isin(
        few_shot_author_ids
    )
].copy()

print(
    f"Candidate authors before exclusion: "
    f"{len(candidate_authors):,}"
)

print(
    f"Candidate authors after exclusion: "
    f"{len(remaining_candidate_authors):,}"
)

print(
    f"Few-shot authors removed: "
    f"{len(candidate_authors) - len(remaining_candidate_authors):,}"
)
```

Candidate authors before exclusion: 24,014
Candidate authors after exclusion: 24,006
Few-shot authors removed: 8

```
In [ ]: evaluation_author_keys = (
    evaluation_authors[["drug_group", "author_id"]]
    .drop_duplicates()
    .assign(evaluation_author=True)
)

extraction_candidate_authors = (
    remaining_candidate_authors
    .merge(evaluation_author_keys, on=["drug_group", "author_id"], how="l
)

extraction_candidate_authors = (
    extraction_candidate_authors[
        extraction_candidate_authors
        ["evaluation_author"].isna()
    ]
)
```

```

        ].drop(columns="evaluation_author").copy()

    )

    print(
        f"Authors after few-shot exclusion: "
        f"{len(remaining_candidate_authors):,}"
    )

    print(
        f"Evaluation authors excluded: "
        f"{len(evaluation_authors):,}"
    )

    print(
        f"Authors remaining for final extraction: "
        f"{len(extraction_candidate_authors):,}"
    )

```

Authors after few-shot exclusion: 24,006
 Evaluation authors excluded: 8
 Authors remaining for final extraction: 23,998

```

In [ ]: few_shot_keys = {
        (example["drug_group"], example["author_id"])
        for example in few_shot_examples
    }

    evaluation_keys = set(
        evaluation_authors[["drug_group", "author_id"]].itertuples(
            index=False, name=None
        )
    )

    extraction_keys = set(
        extraction_candidate_authors[
            ["drug_group", "author_id"]].drop_duplicates().itertuples(
                index=False, name=None
            )
        )

    assert not (few_shot_keys & evaluation_keys), ("Few-shot and evaluation s
    assert not (few_shot_keys & extraction_keys), ("Few-shot and extraction s
    assert not (evaluation_keys & extraction_keys), ("Evaluation and extracti

    print(
        "Confirmed: few-shot, evaluation and "
        "final extraction authors are independent."
    )

```

Confirmed: few-shot, evaluation and final extraction authors are independent.

7. Prompt construction

Schema, consistency rules, inclusion rules, task prompt, and the few-shot block built from the **manually annotated** examples.

```

In [165... extraction_schema = """
{
    "author_id": null,

```

```
"drug_group": null,
"route_of_administration": null,

"include": null,
"exclusion_reason": null,

"longitudinal_criteria": {
  "n_posts": null,
  "days_between_first_and_last_post": null,
  "meets_minimum_requirement": null
},

"treatment_history": {
  "treatment_start_date": null,
  "treatment_start_source": null,
  "reported_treatment_duration": null
},

"drug_switching": {
  "previous_glp1": [],
  "current_glp1": [],
  "reason_for_switch": []
},

"patient_profile": {
  "age_years": null,
  "sex_gender": null,
  "race_ethnicity": null,
  "height_cm": null,

  "baseline_conditions": [
    {
      "specific_label": null,
      "visualisation_group": null
    }
  ]
},

"baseline_metrics": {
  "starting_weight_kg": null,
  "starting_weight_lb": null,
  "starting_bmi": null,
  "goal_weight_kg": null,
  "goal_weight_lb": null
},

"timeline_events": [
  {
    "post_number": null,
    "date": null,

    "treatment_event": {
      "specific_label": null,
      "visualisation_group": null
    },

    "drug": null,
    "dose_mg": null,
    "weight_kg": null,
    "weight_lb": null,
```

```

        "bmi": null,

        "desired_effects": [
            {
                "specific_label": null,
                "visualisation_group": null
            }
        ],

        "adverse_effects": [
            {
                "specific_label": null,
                "visualisation_group": null,
                "severity": null
            }
        ],

        "confounding_variables": [
            {
                "specific_label": null,
                "visualisation_group": null
            }
        ],

        "adherence_or_access_issues": [
            {
                "specific_label": null,
                "visualisation_group": null
            }
        ],

        "explicit_notes": [],
        "inferred_notes": []
    }
],

    "journey_summary": null
}
"""

```

In [166... `schema_consistency_rules = """`

Schema consistency rules:

- Always return every top-level field in the extraction schema.
- Use null for unavailable scalar values.
- Use [] for unavailable list values.
- Never place empty placeholder objects inside arrays.
- previous_glp1, current_glp1 and reason_for_switch must always be arrays.
- Use an empty array when no GLP-1 switch occurred.
- treatment_event must be one object, not an array.
- baseline_conditions, desired_effects, adverse_effects, confounding_variables and adherence_or_access_issues must always be arrays.
- severity must be a string: mild, moderate or severe.
- route_of_administration must be injection, oral, other or null.
- Dates must use YYYY-MM-DD format.

`"""`

In [167... `longitudinal_inclusion_rules = """`

FIRST-PERSON RULE (applies to every post and every extracted item):

- Extract ONLY the author's OWN GLP-1 treatment experience.

- IGNORE any content where the author is advising, replying to, or describing SOMEONE ELSE – e.g. "you should try...", "my wife is on...", "for most nausea fades". Do not create timeline_events, effects, doses, or weight such posts, even when they mention GLP-1 terms.
- A dose, effect, or weight counts ONLY if the author states it about the (first person: "I", "my", "me"). If it is unclear whose experience it is, do NOT extract it.

INCLUSION (whether the journey as a whole is usable):

- include = true only if, using FIRST-PERSON content only, the journey has
 - * at least 2 of the author's own posts spanning at least 14 days, AND
 - * at least one first-person post with treatment-relevant content (dose, weight, effect, adherence, or timeline information).
- include = false if the author never describes their own GLP-1 use (e.g. only give advice, ask questions, or discuss others), or the first-person content is too sparse to meet the thresholds above.

When include = false:

- set exclusion_reason to a short explanation.
 - If excluded because the content is NOT the author's own experience (advice-only, about others, questions-only): extract NOTHING – set effects, events and metrics to empty. Do not record other people's effects.
 - If excluded only because the author's own journey is too short/sparse: you MAY still extract their own desired/adverse effects where clearly present.
- When include = true: set exclusion_reason = null.

"""

```
In [ ]: # Visualisation grouping instructions.
# The vocabulary is derived from the FEW-SHOT annotations only (not the gold)
# so it never leaks gold category names into the gold-standard evaluation
# The model is prompted to prefer these groups, coin a consistent new one
# fits, and only use "other" only as a last resort.

effect_insights = ["desired_effects", "adverse_effects", "confounding_variables",
                  "adherence_or_access_issues"]
semantic_categories = {
    "dose_intolerability": "dose intolerance", "intolerable dosage": "dosage intolerable",
    "dysomnia": "sleep disturbance", "dietary_improvements": "dietary improvements",
    "cholesterol": "high cholesterol", "delayed_scheduled_dose": "delayed scheduled dose",
    "delayed_dose_escalation": "delayed dose escalation",
}
undefined = {"not stated", "side effects", ""}

def build_visualisation_grouping(examples):
    groups = defaultdict(set)

    def add(field, vg):
        if isinstance(vg, str):
            v = vg.strip().lower()
            if v in undefined:
                return
            groups[field].add(semantic_categories.get(v, "other"))

    for ex in examples:
        eo = ex["expected_output"]
        for c in eo.get("patient_profile", {}).get("baseline_conditions"):
            if isinstance(c, dict):
                add("baseline_conditions", c.get("visualisation_group"))
        for te in eo.get("timeline_events") or []:
```

```

        tev = te.get("treatment_event")
        if isinstance(tev, dict):
            add("treatment_event", tev.get("visualisation_group"))
        for fld in effect_insights:
            for it in te.get(fld) or []:
                if isinstance(it, dict):
                    add(fld, it.get("visualisation_group"))

order = ["desired_effects", "adverse_effects", "confounding_variables",
        "adherence_or_access_issues", "treatment_event", "baseline_c
vocab_block = "\n".join(
    f"{cat}:\n  " + "; ".join(sorted(groups[cat]))
    for cat in order if groups[cat]
)

```

```

section = """VISUALISATION GROUPING

```

Every desired effect, adverse effect, confounder, treatment_event, adhere issue and baseline condition must be returned as a SEPARATE item with BOT

- specific_label: the exact issue as reported, concise and lower-case. Ke distinct issue per item – never merge two issues (e.g. reduced food noi reduced cravings) into one item or one label.
- visualisation_group: a broad, chart-ready category, lower-case.

How to choose visualisation_group:

1. PREFER the vocabulary below. If an existing group reasonably fits the use it verbatim – even when the wording in the post differs. Match on not exact words.
2. If NO existing group fits, CREATE a new group, following the naming ru so the same concept gets the same label in any other journey. Do not f to "other" just to avoid naming something.
3. Use "other" ONLY as a last resort – when no clinically or pharmacovigi meaningful label can be applied consistently across journeys (e.g. vag off-topic or uninterpretable content). Prefer a real category whenever

Naming rules for a NEW group (so identical concepts get identical strings independent journeys):

- lower-case; a short noun phrase naming the concept, not a sentence.
- keep all specific detail (symptom, dose, number, body part) in specific NOT in the group name. The group is the reusable bucket.
- do NOT encode severity, dose, frequency, dates or quantifiers in the gr
- follow the directional patterns already in use for effects: "reduced X" / "improved X" / "increased X" / "worsened X".
- for a medical condition or comorbidity, use the recognised clinical ter singular (e.g. "hypothyroidism", "gallbladder disease").
- self-administered supplements -> "supplement use"; prescribed medicatio "co-medication". Never combine the two.
- reuse a group you have already created earlier in THIS journey for the concept, so a journey never contains two spellings of one category.

PREFERRED VOCABULARY (by field):

```

{vocab}"""

```

```

return section.replace("{vocab}", vocab_block)

```

```

visualisation_grouping = build_visualisation_grouping(few_shot_examples)
print(f"visualisation_grouping section: ~{len(visualisation_grouping.spli

```

visualisation_grouping section: ~562 words

In [169...

```
author_journey_prompt = f"""
You extract a structured, longitudinal treatment history for GLP-1 medica

OUTPUT FORMAT
- Return exactly one valid JSON object and nothing else (no markdown, no
- Follow the schema below exactly: same field names, same nesting. Do not
- Unavailable scalar -> null. Unavailable list -> [].
- Do not invent details. Extract only what the text supports, and keep ex

INPUT
- All posts are from ONE author, in chronological order, each with a POST
- drug_group is provided as metadata; use it to label the drug ONLY when

WHAT TO CAPTURE
- The journey over time: drug, dose, weight/BMI, desired effects, adverse
- Timing, taken from the post dates.
- An overall read of whether the evidence suggests efficacy, safety conce

INCLUSION
{longitudinal_inclusion_rules}
Always extract desired and adverse effects even when include = false.

TIMELINE COMPLETENESS
- Produce ONE timeline_events entry for EVERY post that contains any dose
  effect, adverse event, confounder, or adherence/access information.
- Do NOT merge multiple posts into one event, and do NOT summarise the jo
  a few representative events. A long journey must produce many events.
- Copy post_number EXACTLY from the input for each event; never invent po
- Omit a post only when it has no treatment-relevant information.

DRUG & DOSE
- Record drug names as written; map obvious misspellings to the intended
- Label every post with a GLP-1; if only drug_group is known, use that.
- Record non-GLP-1 medications as confounding_variables, never as the tre
- Capture dose transitions as concrete values (e.g. "0.25mg -> 0.5mg"), n
- drug_switching applies ONLY between two GLP-1s (e.g. Ozempic -> Wegovy)

DEMOGRAPHICS, HEIGHT, BMI
- sex_gender: parse forms like 34F / 28M and explicit words (male, female
- height_cm: only if explicitly stated; convert (1 in = 2.54 cm, 1 ft = 3
- Compute bmi = weight_kg / (height_m squared) whenever both are known (1

DATES & DURATION
- reported_treatment_duration only when explicitly quantified ("3 weeks",
- You may infer duration from a stated start plus the post date (e.g. "st
- All dates as YYYY-MM-DD.

CONFOUNDERS & CO-MEDICATIONS
- Extract explicitly mentioned confounders (e.g. PCOS, diabetes, menopaus
- Do NOT exclude an author merely for a comorbidity or a co-medication; r

EFFECTS & SEVERITY
- Desired effects include weight loss, appetite/satiety change, reduced f
- Adverse effects include nausea, vomiting, constipation, diarrhea, reflu
- Whenever an adverse effect is present, set severity to the highest burd

ACTIVITY
- exercise/activity level as low / moderate / high when stated, else null
```

```

{visualisation_grouping}

{schema_consistency_rules}

SCHEMA:
{extraction_schema}

POSTS:
{{text}}

OUTPUT:
"""

```

```

In [ ]: # Build the few-shot block from the MANUALLY ANNOTATED examples.

def build_fewshot_from_annotations(examples):
    blocks = []

    for ex in examples:
        blocks.append(
            f"""POSTS:
{ex['input']}

OUTPUT:
{json.dumps(ex['expected_output'], indent=2, ensure_ascii=False)}"""
        )

    return "\n\n---\n\n".join(blocks)

few_shot = build_fewshot_from_annotations(few_shot_examples)

author_journey_prompt_fewshot = author_journey_prompt.replace(
    "POSTS:\n{text}",
    f"Examples:\n{few_shot}\n\nPOSTS:\n{{text}}"
)

enc = tiktoken.get_encoding("o200k_base")
print(f"Few-shot prompt length: ~{len(enc.encode(author_journey_prompt_fewshot))} tokens")

Few-shot prompt length: ~51,070 tokens (before journey text)

```

8. GPT extraction setup

```

In [ ]: # =====
# GPT setup
# =====

model = "gpt-5.2"
max_output_tokens = 50000 # must also cover reasoning tokens
sleep_time = 0.5
max_retries = 3

# GPT-5.2 accepts temperature/top_p/reasoning_effort; update one, leave the others
# temperature and top_p disabled when reasoning_effort is set.
# temperature = 0
# top_p = 1.0
reasoning_effort = "medium"

```

```

client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])

def parse_model_output(x):
    if x is None:
        return None
    try:
        return json.loads(x)
    except Exception:
        return None

def call_openai(prompt):
    """Single extraction call. JSON output is enforced by the API,
    so parse failures should be rare rather than routine."""

    last_error = None

    for attempt in range(max_retries):
        try:
            response = client.responses.create(
                model=model,
                input=prompt,
                max_output_tokens=max_output_tokens,
                #temperature=TEMPERATURE,
                #top_p=TOP_P,
                #seed=SEED,
                reasoning={"effort": reasoning_effort},
                text={"format": {"type": "json_object"}},
            )
            return response.output_text

        except Exception as e:
            last_error = e
            wait = 2 ** attempt * 5
            print(f" retry {attempt + 1}/{max_retries} after error: {e}")
            time.sleep(wait)

    raise last_error

```

In []: MAX_POSTS_PER_CALL = 60 *# journeys longer than this are extracted in ch*

```

def _chunk_group(group, max_posts=MAX_POSTS_PER_CALL):
    group = group.sort_values("created_utc")
    for start in range(0, len(group), max_posts):
        yield group.iloc[start:start + max_posts]

def _merge_results(parts):
    """Merge per-chunk outputs: concatenate timeline_events, coalesce the
    merged, events = None, []
    for p in parts:
        if not isinstance(p, dict):
            continue
        events.extend(p.get("timeline_events") or [])
        if merged is None:
            merged = {k: v for k, v in p.items() if k != "timeline_events"}
        else: # fill any empty journey fields using chunks
            for k, v in p.items():
                if k != "timeline_events" and merged.get(k) in (None, "",

```

```

        and v not in (None, "", [], {}):
            merged[k] = v
    merged = merged or {}
    # remove duplicate events by post_number, keep chronological order
    seen, uniq = set(), []
    for e in sorted(events, key=lambda e: (e.get("post_number") is None,
        key = e.get("post_number")
        key = json.dumps(e, sort_keys=True) if key is None else key
        if key not in seen:
            seen.add(key); uniq.append(e)
    merged["timeline_events"] = uniq
    return merged

def extract_journey(group, prompt_template, max_posts=MAX_POSTS_PER_CALL)
    """Single call for short journeys; chunked + merged for long ones.
    Returns a JSON string (drop-in for the runner's model_output)."""
    group = group.sort_values("created_utc")
    if len(group) <= max_posts:
        return call_openai(prompt_template.replace("{text}", build_annota
    parts = [parse_model_output(
        call_openai(prompt_template.replace("{text}", build_annot
        for chunk in _chunk_group(group, max_posts)]
    return json.dumps(_merge_results(parts), ensure_ascii=False)

```

```

In [173... def post_coverage(group, model_output):
    parsed = parse_model_output(model_output) if isinstance(model_output,
    if not isinstance(parsed, dict):
        return None
    emitted = {e.get("post_number") for e in (parsed.get("timeline_events
    total = set(group.sort_values("created_utc")["post_number"])
    covered = len(emitted & total)
    return {"n_posts": len(total),
        "n_events": len(parsed.get("timeline_events") or []),
        "posts_covered": covered,
        "coverage": round(covered / len(total), 3) if total else None

```

```

In [ ]: #####
# Extraction runner #
#####

def run_journey_extraction(
    inputs,
    output_path,
    prompt_template,
    dry_run=True,
):
    """
    inputs: list of dicts with keys author_id, drug_group, input (+ metad
           If an item also has a "posts" dataframe, the journey is chunk
           otherwise journey is extracted with one pass from the "input"
    output_path: .jsonl file; one line appended per completed author.
    """

    # Skip authors already present in the output file
    processed = set()
    if os.path.exists(output_path):
        done = pd.read_json(output_path, lines=True)
        if len(done) > 0:
            processed = set(done["author"])

```

```

print(f"Resuming: {len(processed)} authors already processed")

todo = [x for x in inputs if x["author_id"] not in processed]
print(f"Journeys to process: {len(todo)} (of {len(inputs)})")

if dry_run:
    print("DRY RUN – no API calls made")
    return None

with open(output_path, "a", encoding="utf-8") as out_file:

    for n, item in enumerate(todo, start=1):

        # Build the record FIRST, then fill model_output.
        record = {
            # 'author' holds the anonymised author_id; 'drug_group'
            # matches the extraction schema field name.
            "author": item["author_id"],
            "drug_group": item["drug_group"],
            "n_author_posts": item.get("n_author_posts"),
            "first_post": str(item.get("first_post")),
            "last_post": str(item.get("last_post")),
            "days_between_first_and_last_post": item.get("days_between_first_and_last_post"),
            "weeks_between_first_and_last_post": item.get("weeks_between_first_and_last_post"),
            "months_between_first_and_last_post": item.get("months_between_first_and_last_post"),
            "journey_text": item.get("input"),
            "model": model,
            "model_output": None,
            "error": None,
        }

        try:
            if isinstance(item.get("posts"), pd.DataFrame):
                # full extraction: chunk long journeys
                record["model_output"] = extract_journey(item["posts"])
            else:
                record["model_output"] = call_openai(
                    prompt_template.replace("{text}", str(item["input"]))
                )
            print(f"[{n}/{len(todo)}] {item['drug_group']} {item['author_id']}")
        except Exception as e:
            record["error"] = str(e)
            print(f"[{n}/{len(todo)}] {item['drug_group']} {item['author_id']}")

        out_file.write(json.dumps(record, ensure_ascii=False) + "\n")
        out_file.flush()

        time.sleep(sleep_time)

results = pd.read_json(output_path, lines=True)
print(f"\nSaved {len(results)} results to {output_path}")
return results

```

9. Gold-standard Model Evaluation

```

In [178... # Run the model on the gold-standard journeys

evaluation_output_path = f"{gpt_result_dir}/gold_standard_predictions.json"

```

```

dry_run = False    # flip to False to make API calls

evaluation_inputs = [
    {
        "author_id": ex["author_id"],
        "drug_group": ex["drug_group"],
        "input": ex["input"],
    }
    for ex in gold_standard_examples
]

run_journey_extraction(
    inputs=evaluation_inputs,
    output_path=evaluation_output_path,
    prompt_template=author_journey_prompt_fewshot,
    dry_run=dry_run,
)

```

Resuming: 0 authors already processed

Journeys to process: 8 (of 8)

[1/8] oz oz_4a9ba8ff5d43: done

[2/8] oz oz_f06fb96ca50f: done

[3/8] reta reta_bb45854c891d: done

[4/8] reta reta_d0c18b2637e9: done

[5/8] sema sema_46dfbf71f08c: done

[6/8] sema sema_4d9944dbe2bb: done

[7/8] weg weg_2cd38c74d0ef: done

[8/8] weg weg_62ee4e4ed320: done

Saved 8 results to /Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/GPT
_Results/gold_standard_predictions.jsonl

Out [178...

	author	drug_group	n_author_posts	first_post	last_post	days_t
0	oz_4a9ba8ff5d43	oz	NaN	None	None	
1	oz_f06fb96ca50f	oz	NaN	None	None	
2	reta_bb45854c891d	reta	NaN	None	None	
3	reta_d0c18b2637e9	reta	NaN	None	None	
4	sema_46dfbf71f08c	sema	NaN	None	None	
5	sema_4d9944dbe2bb	sema	NaN	None	None	
6	weg_2cd38c74d0ef	weg	NaN	None	None	
7	weg_62ee4e4ed320	weg	NaN	None	None	

```
In [ ]: #####
# Split evaluation: structured fields (exact-match P/R/F1)#
# vs free-text fields (ROUGE-L, BERTScore) #
#####

FREETEXT_KEYS = {"journey_summary", "exclusion_reason", "treatment_start_
                "explicit_notes", "inferred_notes", "specific_label"}
META_KEYS = {"author_id", "drug_group"} # inputs, not predictions -> ex

def _norm(v):
    return re.sub(r"\s+", " ", str(v).strip().lower())

def _flatten(v, parent=""):
    """(leaf_key, path, value) for every non-empty scalar leaf; fixes the
    out = []
    if isinstance(v, dict):
        for k, it in v.items():
            key = k.replace("route_of administration", "route_of_administ
            out += _flatten(it, f"{parent}.{key}" if parent else key)
    elif isinstance(v, list):
```

```

        for it in v:
            out += _flatten(it, parent)
    else:
        nv = _norm(v)
        if nv not in ("", "none", "null", "not reported", "unknown"):
            out.append((parent.split(".")[1], parent.lower(), nv))
    return out

def _structured_facts(obj):
    return {(p, v) for leaf, p, v in _flatten(obj)
            if leaf not in FREETEXT_KEYS and leaf not in META_KEYS}

def _freetext_string(obj):
    return " ".join(v for leaf, p, v in _flatten(obj) if leaf in FREETEXT_KEYS)

def _prf(exp, pred):
    tp = len(exp & pred); fp = len(pred - exp); fn = len(exp - pred)
    p = tp / (tp + fp) if tp + fp else 0.0
    r = tp / (tp + fn) if tp + fn else 0.0
    f = 2 * p * r / (p + r) if p + r else 0.0
    return tp, fp, fn, p, r, f

# ---- load predictions from the checkpoint ----
_preds = pd.read_json(evaluation_output_path, lines=True)
_pred_by = {row["author"]: parse_model_output(row["model_output"])
            for _, row in _preds.iterrows()}

# ---- per-author structured facts + collect prose for batched scoring ----
rows, gold_texts, pred_texts = [], [], []
for ex in gold_standard_examples:
    aid = ex["author_id"]; exp = ex["expected_output"]; pred = _pred_by.get(aid, "")
    E, P = _structured_facts(exp), _structured_facts(pred)
    tp, fp, fn, p, r, f = _prf(E, P)
    rows.append({"author_id": aid, "drug_group": ex["drug_group"],
                "struct_tp": tp, "struct_fp": fp, "struct_fn": fn,
                "struct_precision": round(p, 3), "struct_recall": round(r, 3),
                "struct_f1": round(f, 3)})
    gold_texts.append(_freetext_string(exp) or " ")
    pred_texts.append(_freetext_string(pred) or " ") # avoid empty strings

# ---- free-text: ROUGE-L per author ----
_rs = rouge_scorer.RougeScorer(["rougeL"], use_stemmer=True)
for i, row in enumerate(rows):
    row["freetext_rougeL_f1"] = round(
        _rs.score(gold_texts[i], pred_texts[i])["rougeL"].fmeasure, 3)

# ---- free-text: BERTScore (one call over all authors) ----
_bp, _br, _bf = bert_score(pred_texts, gold_texts, lang="en", rescale_with_baseline=True)
for i, row in enumerate(rows):
    row["freetext_bertscore_f1"] = round(float(_bf[i]), 3)

split_metrics = pd.DataFrame(rows)

# ---- aggregates: macro (mean over authors) + micro (pooled) for structured facts ----
_cols = ["struct_precision", "struct_recall", "struct_f1",
         "freetext_rougeL_f1", "freetext_bertscore_f1"]
macro = split_metrics[_cols].mean().round(3)
TP, FP, FN = (split_metrics[c].sum() for c in ["struct_tp", "struct_fp", "struct_fn"])
micro_p = TP / (TP + FP) if TP + FP else 0.0
micro_r = TP / (TP + FN) if TP + FN else 0.0

```



```

micro_f = 2 * micro_p * micro_r / (micro_p + micro_r) if micro_p + micro_r > 0

# macro mean + std (std shows journey-to-journey variability)
macro_tbl = split_metrics[_cols].agg(["mean", "std"]).round(3).T
macro_tbl.columns = ["macro_mean", "std"]
print("\nMACRO mean +/- std:\n" + macro_tbl.to_string())

# per-drug-group
per_group = split_metrics.groupby("drug_group")[_cols].mean().round(3)
print("\nPer drug_group (indicative, n=2):\n" + per_group.to_string())

print(split_metrics.to_string(index=False))
print("\nMACRO (mean over authors):\n" + macro.to_string())
print(f"\nMICRO structured precision={micro_p:.3f} recall={micro_r:.3f} f1={micro_f:.3f} (TP={TP} FP={FP} FN={FN})")

```

Loading weights: 100%|██████████| 389/389 [00:00<00:00, 6077.93it/s]
[transformers] **RobertaModel** **LOAD REPORT** from: roberta-large

Key	Status
lm_head.dense.bias	UNEXPECTED
lm_head.layer_norm.bias	UNEXPECTED
lm_head.dense.weight	UNEXPECTED
lm_head.layer_norm.weight	UNEXPECTED
lm_head.bias	UNEXPECTED
pooler.dense.bias	MISSING
pooler.dense.weight	MISSING

Notes:

- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

MACRO mean +/- std:

	macro_mean	std
struct_precision	0.664	0.127
struct_recall	0.693	0.079
struct_f1	0.672	0.087
freetext_rougeL_f1	0.243	0.051
freetext_bertscore_f1	0.086	0.089

Per drug_group (indicative, n=2):

	struct_precision	struct_recall	struct_f1	freetext_rougeL_f1
freetext_bertscore_f1				
drug_group				
oz	0.718	0.816	0.764	0.276
0.151				
reta	0.631	0.643	0.620	0.214
0.079				
sema	0.667	0.664	0.662	0.254
0.074				
weg	0.640	0.649	0.644	0.226
0.041				

	author_id	drug_group	struct_tp	struct_fp	struct_fn	struct_precision
struct_recall						
struct_f1						
freetext_rougeL_f1						
freetext_bertscore_f1						
oz_4a9ba8ff5d43	oz	28	9	5		
0.757	0.848	0.800	0.263			0.155
oz_f06fb96ca50f	oz	40	19	11		
0.678	0.784	0.727	0.289			0.147
reta_bb45854c891d	reta	75	97	40		
0.436	0.652	0.523	0.136			-0.048
reta_d0c18b2637e9	reta	19	4	11		
0.826	0.633	0.717	0.291			0.206
sema_46dfbf71f08c	sema	20	6	10		
0.769	0.667	0.714	0.277			0.145
sema_4d9944dbe2bb	sema	35	27	18		
0.565	0.660	0.609	0.232			0.004
weg_2cd38c74d0ef	weg	29	20	17		
0.592	0.630	0.611	0.239			0.039
weg_62ee4e4ed320	weg	22	10	11		
0.688	0.667	0.677	0.214			0.043

MACRO (mean over authors):

struct_precision	0.664
struct_recall	0.693
struct_f1	0.672
freetext_rougeL_f1	0.243
freetext_bertscore_f1	0.086

MICRO structured precision=0.583 recall=0.685 f1=0.630 (TP=268 FP=192 FN=123)

```
In [ ]: split_metrics.to_csv(  
        f"{test_output_dir}/evaluation_metrics_by_author.csv",  
        index=False,  
    )  
  
    # macro_tbl is indexed by metric name  
    macro_tbl.to_csv(  
        f"{test_output_dir}/evaluation_metrics_macro.csv"  
    )  
  
    # micro scores are scalars in the split cell
```

```

micro_results = pd.DataFrame({
    "metric": ["micro_precision", "micro_recall", "micro_f1"],
    "score": [round(micro_p, 3), round(micro_r, 3), round(micro_f, 3)],
})
micro_results.to_csv(
    f"{test_output_dir}/evaluation_metrics_micro.csv",
    index=False,
)

# per_group is indexed by drug_group
per_group.to_csv(
    f"{test_output_dir}/evaluation_metrics_by_drug_group.csv"
)

print("Evaluation metrics saved.")

```

Evaluation metrics saved.

10. Checking full extraction outputs

```

In [ ]: final_extraction_journeys = (
    longitudinal_pool
    .merge(extraction_candidate_authors[["drug_group", "author_id"]].drop_duplicates(
        on=["drug_group", "author_id"], how="inner")
    .sort_values(["drug_group", "author_id", "created_utc"]).reset_index()
)

print(
    f"Final extraction authors: "
    f"{final_extraction_journeys[['drug_group', 'author_id']].drop_duplicates().shape[0]}
)

print(
    f"Final extraction posts: "
    f"{len(final_extraction_journeys):,}"
)

```

Final extraction authors: 23,998
 Final extraction posts: 1,223,215

```

In [151]: print(pd.DataFrame(llm_inputs)["n_author_posts"].describe())

```

```

count    23998.000000
mean         50.971539
std        160.088769
min           2.000000
25%          8.000000
50%         19.000000
75%         47.000000
max        8794.000000
Name: n_author_posts, dtype: float64

```

```

In [152]: n = pd.DataFrame(llm_inputs)["n_author_posts"]
print(n.describe(percentiles=[.9, .95, .99, .999]))
print("journeys > 100 posts:", (n > 100).sum())
print("journeys > 300 posts:", (n > 300).sum())

```

```

count      23998.000000
mean        50.971539
std         160.088769
min          2.000000
90%         107.000000
95%         178.000000
99%         467.060000
99.9%       2197.045000
max         8794.000000
Name: n_author_posts, dtype: float64
journeys > 100 posts: 2593
journeys > 300 posts: 515

```

```

In [ ]: llm_inputs = []

for (drug_group, author_id), group in final_extraction_journeys.groupby(
    ["drug_group", "author_id"], sort=False
):
    group = group.sort_values("created_utc")
    first_post = group["created_utc"].min()
    last_post = group["created_utc"].max()
    days_between = (last_post - first_post).days

    llm_inputs.append({
        "author_id": author_id,
        "drug_group": drug_group,
        "n_author_posts": len(group),
        "first_post": first_post,
        "last_post": last_post,
        "days_between_first_and_last_post": days_between,
        "weeks_between_first_and_last_post": round(days_between / 7, 1),
        "months_between_first_and_last_post": round(days_between / 30.44,
        "posts": group, # dataframe for chunking
        "input": build_annotation_input(group), # for dry-run/preview on
    })

print(f"Prepared {len(llm_inputs):,} author journeys for LLM extraction")

```

Prepared 23,998 author journeys for LLM extraction

```

In [ ]: #####
# Select 450 randomly sampled authors per drug group for extraction #
#####

TARGET_PER_GROUP = 450
SEED = 42

inputs_df = pd.DataFrame(llm_inputs)

counts = inputs_df["drug_group"].value_counts()
print("available authors per drug_group:")
print(counts.to_string())

# sample per group WITHOUT losing drug_group: select columns explicitly a
sampled_df = (
    inputs_df
    .groupby("drug_group", group_keys=False)[inputs_df.columns]
    .apply(lambda g: g.sample(n=min(TARGET_PER_GROUP, len(g)), random_sta
    .reset_index(drop=True)
)

```

```
assert "drug_group" in sampled_df.columns, "drug_group lost in sampling"

extraction_inputs = sampled_df.to_dict("records")

print(f"\nselected {len(extraction_inputs)} authors:")
print(sampled_df["drug_group"].value_counts().sort_index().to_string())
print("drug_group on first item:", "drug_group" in extraction_inputs[0])

# ---- run the extraction ----
scaled_output_path = f"{gpt_result_dir}/extraction_450_per_group.jsonl"

dry_run = False # flip True for dry run or False to make API calls

gpt_results = run_journey_extraction(
    inputs=extraction_inputs,
    output_path=scaled_output_path,
    prompt_template=author_journey_prompt_fewshot,
    dry_run=dry_run,
)
```

available authors per drug_group:

drug_group

sema	9075
oz	8526
reta	5874
weg	523

selected 1800 authors:

drug_group

oz	450
reta	450
sema	450
weg	450

drug_group on first item: True

Journeys to process: 1800 (of 1800)

[1/1800] oz oz_e1033af173bb: done
[2/1800] oz oz_583a6a9c7693: done
[3/1800] oz oz_fcd929b9e873: done
[4/1800] oz oz_20971e1af8e1: done
[5/1800] oz oz_19d743b3d8c1: done
[6/1800] oz oz_7eba97e3fb92: done
[7/1800] oz oz_3fd577c8fe60: done
[8/1800] oz oz_2905e0a36a43: done
[9/1800] oz oz_b840a6d9afbe: done
[10/1800] oz oz_c2b9f79634d1: done
[11/1800] oz oz_85567763fae0: done
[12/1800] oz oz_313296e5ce9c: done
[13/1800] oz oz_7ef8922b84ef: done
[14/1800] oz oz_2fb1535e27e0: done
[15/1800] oz oz_9c0c5b5e25ed: done
[16/1800] oz oz_953acd5a6cda: done
[17/1800] oz oz_0922c4918015: done
[18/1800] oz oz_69210be19541: done
[19/1800] oz oz_e766cb0cebd2: done
[20/1800] oz oz_3e720ffeabc38: done
[21/1800] oz oz_83ef59c0f2ea: done
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[1554/1800] weg weg_cd92c7557a89: done
[1555/1800] weg weg_f4464d386fed: done
[1556/1800] weg weg_a96963a1a534: done
[1557/1800] weg weg_e15d022fec68: done
[1558/1800] weg weg_416f4995937e: done
[1559/1800] weg weg_b8b47f3887b5: done
[1560/1800] weg weg_22a12e9e62e7: done
[1561/1800] weg weg_f1907d0c1c2b: done
[1562/1800] weg weg_52056eb47ef7: done
[1563/1800] weg weg_5eadd0ad8cbe: done
[1564/1800] weg weg_f342c1144a8b: done
[1565/1800] weg weg_be2191519a38: done
[1566/1800] weg weg_777d9f1ff1a2: done
[1567/1800] weg weg_5ff8f24b1a80: done
[1568/1800] weg weg_28b392442d90: done
[1569/1800] weg weg_33b695a128e2: done
[1570/1800] weg weg_fcd8c53603d8: done
[1571/1800] weg weg_438624f583a7: done
[1572/1800] weg weg_2dc9a6e0a22f: done
[1573/1800] weg weg_47dd197f96ff: done
[1574/1800] weg weg_7ae05adce97a: done
[1575/1800] weg weg_c6cde6fa18c3: done
[1576/1800] weg weg_f945027e1bd4: done
[1577/1800] weg weg_29ee79caf3e9: done
[1578/1800] weg weg_317160e78788: done
[1579/1800] weg weg_4f23145a8c08: done
[1580/1800] weg weg_4e3c1966de57: done
[1581/1800] weg weg_61b4c0463839: done
[1582/1800] weg weg_7ea1232ccd0a: done
[1583/1800] weg weg_24bf81c0a8d0: done
[1584/1800] weg weg_9e2d6641f748: done
[1585/1800] weg weg_77c489a3ba27: done
[1586/1800] weg weg_f3978394f799: done
[1587/1800] weg weg_36e805b9aa82: done
[1588/1800] weg weg_f6807236fc13: done
[1589/1800] weg weg_99a60f162b03: done
[1590/1800] weg weg_82d748562bd0: done
[1591/1800] weg weg_b4ab9cffeef3: done
[1592/1800] weg weg_d90d1b4ff974: done
[1593/1800] weg weg_f1a9731299cf: done
[1594/1800] weg weg_960f8d156152: done
[1595/1800] weg weg_bbf33234c5c2: done
[1596/1800] weg weg_f9a3183db30d: done
[1597/1800] weg weg_d0942a0e276f: done
[1598/1800] weg weg_a54806fe115d: done
[1599/1800] weg weg_a4101d8d3e29: done
[1600/1800] weg weg_108e0949ac4d: done
[1601/1800] weg weg_60c97b89d67b: done
[1602/1800] weg weg_3ed986591242: done
[1603/1800] weg weg_b84c37431e6a: done
[1604/1800] weg weg_8261035af7da: done
[1605/1800] weg weg_9384ddcf1138: done

[1606/1800] weg weg_93b0c9ed6b66: done
[1607/1800] weg weg_208b76f3bbf9: done
[1608/1800] weg weg_33f9fb77fd97: done
[1609/1800] weg weg_8926cbd8662a: done
[1610/1800] weg weg_849ab9070de2: done
[1611/1800] weg weg_b66ae0ebe2ce: done
[1612/1800] weg weg_45edd873ec18: done
[1613/1800] weg weg_c20fc42f514c: done
[1614/1800] weg weg_1280de9d324b: done
[1615/1800] weg weg_fd864015e19e: done
[1616/1800] weg weg_6139aa4d1126: done
[1617/1800] weg weg_cdbbf0c00659: done
[1618/1800] weg weg_43becf61c6c1: done
[1619/1800] weg weg_c6a3f35c2756: done
[1620/1800] weg weg_4457423aef50: done
[1621/1800] weg weg_9d62acaccfd5: done
[1622/1800] weg weg_93d89ebe6ec9: done
[1623/1800] weg weg_38fd9735f55a: done
[1624/1800] weg weg_dcd15023315a: done
[1625/1800] weg weg_2f0cced31c2e: done
[1626/1800] weg weg_42276edcd730: done
[1627/1800] weg weg_77d4c12643e9: done
[1628/1800] weg weg_8bdf38b5d676: done
[1629/1800] weg weg_2f6e93ec3ac2: done
[1630/1800] weg weg_c839276249ef: done
[1631/1800] weg weg_37e8a592393c: done
[1632/1800] weg weg_5d02f182d454: done
[1633/1800] weg weg_62bcf4f82351: done
[1634/1800] weg weg_7bb77117b7fe: done
[1635/1800] weg weg_e4d9b4ffb6e5: done
[1636/1800] weg weg_989828326fd4: done
[1637/1800] weg weg_b2faf73a8fd1: done
[1638/1800] weg weg_9a160ed24eaf: done
[1639/1800] weg weg_c81ae7bea1ee: done
[1640/1800] weg weg_cb31490346ed: done
[1641/1800] weg weg_39b9a6a2c8f4: done
[1642/1800] weg weg_e04260d3a02d: done
[1643/1800] weg weg_6d8c7420ac81: done
[1644/1800] weg weg_d3990cf607ad: done
[1645/1800] weg weg_6ab7913b31fd: done
[1646/1800] weg weg_3b7962c5f8f6: done
[1647/1800] weg weg_d63102e01abe: done
[1648/1800] weg weg_c4dd32464a38: done
[1649/1800] weg weg_aa25f2cb5422: done
[1650/1800] weg weg_be8739b3cd06: done
[1651/1800] weg weg_9551e8fe3d47: done
[1652/1800] weg weg_b4d26a0e26b0: done
[1653/1800] weg weg_9b8d29d21c9b: done
[1654/1800] weg weg_918590ea61f4: done
[1655/1800] weg weg_b8f90d502d55: done
[1656/1800] weg weg_af8429dd1331: done
[1657/1800] weg weg_349b7d24576a: done
[1658/1800] weg weg_5825a5ce5e38: done
[1659/1800] weg weg_9e2b439eb3d1: done
[1660/1800] weg weg_af2e33e53055: done
[1661/1800] weg weg_c40189754aeb: done
[1662/1800] weg weg_fa4adb09d771: done
[1663/1800] weg weg_b642f39ed286: done
[1664/1800] weg weg_75d1f14985d8: done
[1665/1800] weg weg_e33105603337: done

[1666/1800] weg weg_a10e5cd0a6d5: done
[1667/1800] weg weg_4e94d91e2463: done
[1668/1800] weg weg_9c44f031728a: done
[1669/1800] weg weg_3e1c84fe0085: done
[1670/1800] weg weg_60e078b49a66: done
[1671/1800] weg weg_80f1e120020e: done
[1672/1800] weg weg_7473151353d1: done
[1673/1800] weg weg_36209c1df839: done
[1674/1800] weg weg_373f8f5ce5f6: done
[1675/1800] weg weg_b176bab6dc97: done
[1676/1800] weg weg_6e9498c861fe: done
[1677/1800] weg weg_cb450cf61349: done
[1678/1800] weg weg_b015f9cf8c7e: done
[1679/1800] weg weg_3b078480c328: done
[1680/1800] weg weg_8cd4e9c26ef2: done
[1681/1800] weg weg_cf7eff779c22: done
[1682/1800] weg weg_3368912fd5be: done
[1683/1800] weg weg_d6f01730ef24: done
[1684/1800] weg weg_dc9893f3eb68: done
[1685/1800] weg weg_3d4a49066753: done
[1686/1800] weg weg_b825560928fd: done
[1687/1800] weg weg_149cb8f9dda0: done
[1688/1800] weg weg_eb9cf6698d88: done
[1689/1800] weg weg_22598a1cad34: done
[1690/1800] weg weg_8ba3598d52e1: done
[1691/1800] weg weg_2ba3dec287a1: done
[1692/1800] weg weg_7a7d0aab1c2e: done
[1693/1800] weg weg_5e1063815861: done
[1694/1800] weg weg_c3eda9f3178f: done
[1695/1800] weg weg_4b8b499cfe66: done
[1696/1800] weg weg_093596adef33: done
[1697/1800] weg weg_102b13d0d0f7: done
[1698/1800] weg weg_0ceac97c8e03: done
[1699/1800] weg weg_53001afae631: done
[1700/1800] weg weg_41bfa036bfa4: done
[1701/1800] weg weg_c8f167c3bf23: done
[1702/1800] weg weg_afd3f7abf262: done
[1703/1800] weg weg_6d33c3f3fa91: done
[1704/1800] weg weg_2f0b58df3afb: done
[1705/1800] weg weg_1d24a7035bb0: done
[1706/1800] weg weg_78c0a9fcaa45: done
[1707/1800] weg weg_d60145a8178e: done
[1708/1800] weg weg_ae163eb2db0b: done
[1709/1800] weg weg_c0b78f6bab51: done
[1710/1800] weg weg_57cb241d8c05: done
[1711/1800] weg weg_13439bf5b88f: done
[1712/1800] weg weg_dd949dc59e4b: done
[1713/1800] weg weg_c4e24e862af2: done
[1714/1800] weg weg_6485c0a37006: done
[1715/1800] weg weg_8b2ee919f335: done
[1716/1800] weg weg_b023536b9429: done
[1717/1800] weg weg_7e3b6c078cf7: done
[1718/1800] weg weg_cff531848c2a: done
[1719/1800] weg weg_04353317ce0f: done
[1720/1800] weg weg_7fb25824231e: done
[1721/1800] weg weg_e3f2e170304b: done
[1722/1800] weg weg_2ff226ed2e7c: done
[1723/1800] weg weg_6fd4accb26fa: done
[1724/1800] weg weg_d9e1ff5bc27b: done
[1725/1800] weg weg_688a311e38f5: done

[1726/1800] weg weg_d875418ee37a: done
[1727/1800] weg weg_5332eb6d8c66: done
[1728/1800] weg weg_2f85975f0cbb: done
[1729/1800] weg weg_9012269ec5fc: done
[1730/1800] weg weg_693fe8d92ee5: done
[1731/1800] weg weg_20bcbcb17f3cc: done
[1732/1800] weg weg_1714465359e1: done
[1733/1800] weg weg_0fc39c3bf4b6: done
[1734/1800] weg weg_8413c0136369: done
[1735/1800] weg weg_a211a68f5258: done
[1736/1800] weg weg_61bd20b3a7db: done
[1737/1800] weg weg_e30e58ffe39f: done
[1738/1800] weg weg_0c5c6e265c8c: done
[1739/1800] weg weg_c57265ee3ddb: done
[1740/1800] weg weg_734b76c7c389: done
[1741/1800] weg weg_821b097c392d: done
[1742/1800] weg weg_b683f1ea8041: done
[1743/1800] weg weg_8ec4c64d05f9: done
[1744/1800] weg weg_4e0dea62b915: done
[1745/1800] weg weg_bd2503b2b5a2: done
[1746/1800] weg weg_e722f74a33d6: done
[1747/1800] weg weg_3e88c2e61939: done
[1748/1800] weg weg_21e517e547e2: done
[1749/1800] weg weg_3e1489d08eec: done
[1750/1800] weg weg_3b28ac3e22cf: done
[1751/1800] weg weg_f80951b6dc13: done
[1752/1800] weg weg_058b0b40fb09: done
[1753/1800] weg weg_a1f16a278793: done
[1754/1800] weg weg_ee51e12d0487: done
[1755/1800] weg weg_224d52fd9d7c: done
[1756/1800] weg weg_949759f9a61b: done
[1757/1800] weg weg_095a1ef051e7: done
[1758/1800] weg weg_4a9591292482: done
[1759/1800] weg weg_130c7e88a058: done
[1760/1800] weg weg_b6a20e22a51d: done
[1761/1800] weg weg_e9fa599530c8: done
[1762/1800] weg weg_cb9717a601d7: done
[1763/1800] weg weg_6a0e854e1865: done
[1764/1800] weg weg_c5c721d1eb42: done
[1765/1800] weg weg_8ac51a12c3d8: done
[1766/1800] weg weg_dd7350f50aba: done
[1767/1800] weg weg_dbd0687ba9cb: done
[1768/1800] weg weg_a7d69edccccc: done
[1769/1800] weg weg_76abb515387d: done
[1770/1800] weg weg_6582e3698956: done
[1771/1800] weg weg_67eaa818d903: done
[1772/1800] weg weg_91ec14d0e2d8: done
[1773/1800] weg weg_e79620d60632: done
[1774/1800] weg weg_7d7e7aef669f: done
[1775/1800] weg weg_fd414ef90bf4: done
[1776/1800] weg weg_e8e985039569: done
[1777/1800] weg weg_95d3deaf1c1d: done
[1778/1800] weg weg_adc59dfc6989: done
[1779/1800] weg weg_84bcbf3781ee: done
[1780/1800] weg weg_628596deca24: done
[1781/1800] weg weg_4d3b7f5faac8: done
[1782/1800] weg weg_1408d25bbeb0: done
[1783/1800] weg weg_6a9269e77b04: done
[1784/1800] weg weg_c393b8eb7b65: done
[1785/1800] weg weg_5f22a6ced880: done

```

[1786/1800] weg weg_9850b9d26798: done
[1787/1800] weg weg_816131565457: done
[1788/1800] weg weg_31d047f6878c: done
[1789/1800] weg weg_1ee06fefa5d6: done
[1790/1800] weg weg_bbcd70b100ce: done
[1791/1800] weg weg_014a2999ca9e: done
[1792/1800] weg weg_e9039680bc9f: done
[1793/1800] weg weg_face14096b43: done
[1794/1800] weg weg_18f3a857d510: done
[1795/1800] weg weg_c95585468d86: done
[1796/1800] weg weg_289a8a3fcb61: done
[1797/1800] weg weg_643de9045798: done
[1798/1800] weg weg_0fe3c95e7874: done
[1799/1800] weg weg_ceef7df2245e: done
[1800/1800] weg weg_8297299c3b5e: done

```

Saved 1800 results to /Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/GPT_Results/extraction_450_per_group.jsonl

```

In [ ]: #####
# Post-coverage monitor for the full extraction #
#####

COVERAGE_FLAG = 0.70 # journeys below this are flagged for review

# content posts = posts that passed the seed-relevance screen
_content = (
    final_extraction_journeys[final_extraction_journeys["relevance"]]
    .groupby(["drug_group", "author_id"])["post_number"]
    .apply(set)
    .to_dict()
)
# fall back to all posts if an author has no relevant rows
# coverage against ALL posts (relevance under-counts content, esp. for re
_allposts = (
    final_extraction_journeys
    .groupby(["drug_group", "author_id"])["post_number"]
    .apply(set).to_dict()
)

cov_rows = []
for _, r in _scaled.iterrows():
    parsed = parse_model_output(r["model_output"])
    aid, dg = r["author"], r["drug_group"]
    allposts = _allposts.get((dg, aid), set())
    events = (parsed or {}).get("timeline_events") or []
    emitted = {e.get("post_number") for e in events if isinstance(e, dict)}
    covered = len(emitted & allposts)
    cov_rows.append({
        "author_id": aid, "drug_group": dg,
        "n_posts": len(allposts),
        "n_events": len(events),
        "posts_covered": covered,
        "coverage": round(covered / len(allposts), 3) if allposts else No
        "valid_json": parsed is not None,
    })
coverage_df = pd.DataFrame(cov_rows)

# summary
print(f"journeys: {len(coverage_df)}")

```

```

print(f"invalid/empty JSON: {(~coverage_df['valid_json']).sum()}")
print(f"median coverage: {coverage_df['coverage'].median():.2f}")
print(f"flagged (< {COVERAGE_FLAG:.0%}): {(coverage_df['coverage'] < COVE

# is coverage worse on longer journeys? (the length-bias check)
print("\ncoverage by journey length:")
print(
    coverage_df.assign(
        length_bucket=pd.cut(coverage_df["n_posts"],
                               [0, 10, 30, 60, 10_000],
                               labels=["1-10", "11-30", "31-60", "60+"])
    )
    .groupby("length_bucket", observed=True)["coverage"]
    .agg(["count", "median"]).round(3).to_string()
)

# the worst offenders to eyeball / re-extract
flagged = coverage_df[coverage_df["coverage"] < COVERAGE_FLAG].sort_value
print(f"\nlowest-coverage journeys:\n{flagged.head(15).to_string(index=Fa

coverage_df.to_csv(f"{gpt_result_dir}/full_extraction_coverage.csv", inde

```

```

journeys: 1800
invalid/empty JSON: 0
median coverage: 0.53
flagged (< 70%): 1278

```

```

coverage by journey length:
              count  median
length_bucket
1-10           659    0.750
11-30          561    0.571
31-60          262    0.423
60+           318    0.333

```

```

lowest-coverage journeys:
      author_id drug_group  n_posts  n_events  posts_covered  coverage
valid_json
reta_ad87e7ce0df7      reta         3         0             0         0.0
True
reta_65e21ddd07df      reta        14         0             0         0.0
True
  weg_6139aa4d1126      weg         13         0             0         0.0
True
  weg_c8f167c3bf23      weg          8         0             0         0.0
True
sema_cf1b8ab997fa      sema        31         0             0         0.0
True
sema_ca4a185f2a0d      sema        11         0             0         0.0
True
sema_1689902c87a4      sema          9         0             0         0.0
True
  oz_2c1e6a4f9b6a       oz         14         0             0         0.0
True
sema_67ffff2ca0cf      sema          9         0             0         0.0
True
reta_6c123957e936      reta        16         0             0         0.0
True
reta_0f8993cab34b      reta          4         0             0         0.0
True
reta_ea4d40a1fcf7      reta        13         0             0         0.0
True
reta_ac83ded1809e      reta        10         0             0         0.0
True
reta_dc09b6acbcf8      reta          5         0             0         0.0
True
reta_01163194ed94      reta          3         0             0         0.0
True

```

```

In [ ]: # Checking why events show empty on both include=false (correct) and genu
zero_ev = coverage_df[coverage_df["n_events"] == 0]
print(f"{len(zero_ev)} journeys with 0 events")
_by = {r["author"]: parse_model_output(r["model_output"]) for _, r in _sc
for aid in zero_ev["author_id"].head(15):
    p = _by.get(aid) or {}
    print(f"{aid}: include={p.get('include')} | reason={p.get('exclusion_

```

64 journeys with 0 events
 oz_d56dc040c3d1: include=False | reason=Posts discuss Medicare/policy coverage for Wegovy and other medications but do not describe the author's own GLP-1 treatment use (no first-person dosing, effects, weight, or adherence history).
 oz_382b3f3d51c8: include=False | reason=No clear first-person GLP-1 treatment use, dosing, effects, weight, or adherence information; posts are primarily advice/commentary about others.
 oz_c3f33f946118: include=False | reason=author does not describe their own GLP-1 use; posts are advice/general information only
 oz_1cf9dd669a2d: include=False | reason=author does not describe their own glp-1 use; posts are advice/questions about others
 oz_aeea4e9f7c1e: include=False | reason=No first-person GLP-1 medication use, dosing, effects, weight/BMI, or access/adherence details were described; posts are general advice about exercise/body composition.
 oz_ac3b8cc00753: include=False | reason=Author does not describe their own GLP-1 medication use; posts are questions/advice and they explicitly state they have never been on ozempic.
 oz_2c1e6a4f9b6a: include=False | reason=No first-person GLP-1 treatment use described; posts are about considering treatment, access, or advising others.
 reta_ea4d40a1fcf7: include=True | reason=None
 reta_01163194ed94: include=False | reason=Author mentions retatrutide but provides no first-person treatment details (no dose, weight, effects, adherence, or timing).
 reta_91f65777b366: include=False | reason=No first-person GLP-1 treatment use described; posts are about planned dosing/questions before starting.
 reta_34523cff2998: include=False | reason=author does not describe their own glp-1 medication use; posts discuss trt and general commentary
 reta_b8a8c962b8cd: include=False | reason=no clear first-person GLP-1 treatment experience; posts are primarily general discussion/advice about retatrutide and other GLP-1s
 reta_2745e174ecd3: include=False | reason=No first-person GLP-1 medication use described; posts are primarily advice/opinions about dosing and safety.
 reta_c442af0dad98: include=False | reason=author does not describe their own GLP-1 medication use; posts are advice/discussion without first-person GLP-1 treatment details
 reta_ada88e7988ef: include=False | reason=Posts discuss GLP-1 dosing safety in general/for others; author does not describe their own GLP-1 use.

```
In [209... inc = coverage_df.merge(
    pd.DataFrame([{"author_id": a, "include": (_by[a] or {}).get("include", "author_id")
  ])
  inc = inc[inc["include"] == True].copy()
  inc["events_per_post"] = inc["n_events"] / inc["n_posts"].clip(lower=1)
  print(inc.groupby(pd.cut(inc["n_posts"], [0,10,30,60,10000]))["events_per
```

```
n_posts
(0, 10]      0.777778
(10, 30]     0.583333
(30, 60]     0.428571
(60, 10000]  0.343750
Name: events_per_post, dtype: float64
```

```
In [210... long_low = inc[(inc["n_posts"] > 60)].sort_values("events_per_post").iloc
aid = long_low["author_id"]; dg = long_low["drug_group"]
posts = final_extraction_journeys[
    (final_extraction_journeys["author_id"]==aid) & (final_extraction_jou
```

```
].sort_values("created_utc")
print(f"{aid}: {len(posts)} posts -> {long_low['n_events']} events\n")
for _, row in posts.head(30).iterrows():
    print("•", row["body"][:150].replace("\n", " "))
# then compare against:
print("\nEVENTS:")
for e in (_by[aid].get("timeline_events") or []):
    print(" -", [d.get("specific_label") for d in (e.get("desired_effects")
        [a.get("specific_label") for a in (e.get("adverse_effects"))])])
```


reta_e69eb1b18052: 67 posts -> 1 events

- Holding more fat in your chest isn't a sign of high estrogen it's juust where he holds a lot of fat. it's no different from holding a lot of fat i n th
- Sounds like you're nutrient deficient
- Tren ace EOD and fewer sides and mood stability shouldn't be in the same sentence, ace should be done ED
- bro what, you adjust the dose so you do less ED. Its not cleaner on pape r its literally cleaner in your blood. The half life is 24-48 hours. If yo u ca
- 2 days isn't enough time to show sustained weight loss, less food will r esult in more weight loss.
- >stubborn abdomen and hip Im sorry but its my understanding that tesam orelin only helps reduce visceral fat. Abdomen and hip fat would fall un der
- You need to decrease your calories if you want more weight loss, stalls aren't dose related. Reta will slightly increase your tdee but if you los e en
- You either have a metabolic disease like inactive thyroid or you are cou nting wrong. Do you weigh everything you eat ? Every time you cook with oi l do
- Bro the only reason you don't know what the long term side effects from cjc/ipa is because no study has made it past few months before getting can cell
- Literally none of this is evidence based and is just assumptions and fee lings. Hyperplasia only happens with crazy dosages 20iu+. All of the side effe
- Hgh is a peptide why do you keep saying peptides stimulate production. It's like you are confusing hgh with mk or something. You take 1 shot of hgh
- You don't cycle tb500, you use it while healing whatever injury you are trying to heal then you don't use it. There is no reason to use bpc or t b if
- You probably shouldn't use Klow if you don't have an injury. Klow makes no sense these peptides have different use cases. Klow/glow is just clever mar
- There isn't any evidence to the contrary either, but because it increase s VEGFR2 → AKT/eNOS signaling. More blood vessels can theoretically feed t umor
- Like I said I just said it theoretically could, not saying it's 100% def initive. I'd like it if you could answer my other questions though, I genu inel
- Why do you think it will peak on day 5/6, when at that time it should've halfed not peaked. half life of reta is around 6 days as in half of it ha s b
- It doesn't make sense pharmacologically though, even if someone were a s low metaboliser of the drug it wouldn't last 2 or 3x as long in the body.
- Yeah maybe for you but I highly doubt it because there literally isn't a documented case of this happening in any of the studies. You shouldn't p res
- Giving false information is worse than no information, also I corrected your false information. There are published studies on reta, just becaus e wh
- Shutting down natural test production to cut on rad is crazy work
- Triz raises rhr, Reta rises it a bit more, but it still does
- 500cals above maintenance is crazy overkill, 200 will achieve the same w ith less fat gain
- you don't need to gain 90% fat for every 10% muscle you gain
- This is what Janoshik recommends. • Room temp: Years for properly made

lyophilized vials • Fridge: Years of viability • Freezer: Potentially DECADE

- Yeah like a 100–150 more calories maybe, a natural doesn't need to eat above 100–150 calories above maintenance. 500 calories above maintenance is 2.2
- Sorry I wasn't clear an enhanced lifter doesn't need to eat in a surplus that's more than 300 calories, a natural lifter shouldn't eat more than a 10
- Used to take tesa 2mg every night for around 3 months didn't really see any improvement just more water retention. Switched to Hgh and it's been way better
- I think it's safe to assume most people aren't getting a bad reaction from their injections. I pin daily both IM and subq and it doesn't feel like I'm
- Animal based diet is high in saturated fats which is counterproductive if you want to keep LDL, and general heart markers in range
- I'll listen to cardiologist over random people on Reddit or psychiatrist trying to sell me on some alternative diet.

EVENTS:

- ['not hungry while on it'] []

```
In [ ]: #####
# ANALYSIS SET #
#####

MIN_FIRST_PERSON_EVENTS = 2      # number of extracted events required
MIN_DESIRED_OR_ADVERSE = 1      # number of adverse and desired effects r

def summarise(aid, p):
    p = p or {}
    events = p.get("timeline_events") or []
    n_events = sum(1 for e in events if isinstance(e, dict))
    n_desired = sum(len(e.get("desired_effects") or []) for e in events if
    n_adverse = sum(len(e.get("adverse_effects") or []) for e in events if
    return {
        "author_id": aid,
        "include_model": p.get("include"),
        "n_events": n_events,
        "n_desired": n_desired,
        "n_adverse": n_adverse,
        "n_effects": n_desired + n_adverse,
    }

analysis = pd.DataFrame([summarise(a, _by[a]) for a in _by])
analysis = analysis.merge(
    _scaled[["author", "drug_group", "n_author_posts"]].rename(columns={"
    on="author_id", how="left",
)

keep = (
    (analysis["include_model"] == True) &
    (analysis["n_events"] >= MIN_FIRST_PERSON_EVENTS) &
    (analysis["n_effects"] >= MIN_DESIRED_OR_ADVERSE)
)
analysis["in_analysis_set"] = keep

print("full extraction:", len(analysis))
print(f"model include=True: {(analysis['include_model']==True).sum()}")
print(f"in analysis set (>= {MIN_FIRST_PERSON_EVENTS} events, >= {MIN_DES
```

```

print("\nanalysis set by drug_group:")
print(analysis[keep].groupby("drug_group").size().to_string())
print("\ndropped by reason:")
print(f"  model-excluded:      {(analysis['include_model'] != True).sum}")
print(f"  too few events:      {((analysis['include_model']==True) & (a
print(f"  no effects extracted:  {((analysis['include_model']==True) & (a

analysis_set_ids = set(analysis[keep]["author_id"])
analysis.to_csv(f"{gpt_result_dir}/analysis_set_membership.csv", index=Fa
print(f"\nanalysis set: {len(analysis_set_ids)} authors -> use these for

```

full extraction: 1800
 model include=True: 1688
 in analysis set (>= 2 events, >= 1 effect): 1633

analysis set by drug_group:

```

drug_group
oz      422
reta    392
sema    409
weg     410

```

dropped by reason:

```

  model-excluded:      112
  too few events:      39
  no effects extracted: 16

```

analysis set: 1633 authors -> use these for aggregate stats

11. Interpretation and visualisation

Cohort EDA

```

In [ ]: #####
# Cohort Selection EDA Setup #
#####

# ---- drug display constants (shared by every figure) ----
DRUG_COLOUR = {"reta": "#e41a1c", "sema": "#377eb8", "weg": "#7c4aaf", "o
DRUG_NAME    = {"reta": "Retatrutide", "sema": "Semaglutide", "weg": "Wego
DRUG_ORDER   = ["reta", "sema", "weg", "oz"]
SEMA_FAMILY  = ("oz", "sema", "weg")          # pooled semaglutides

def wilson(reporters, total, z=1.96):
    """
    Wilson score 95% confidence interval for a proportion.
    reporters = number who reported the effect (k)
    total     = number of authors in the group (sample size, n)
    z         = 1.96 gives a 95% interval
    Returns (rate, ci_low, ci_high). The Wilson interval is used instead
    of the simple normal approximation because it stays inside [0,1] and
    performs well in small samples and rare effects.
    """
    if total == 0:
        return (np.nan, np.nan, np.nan)
    rate      = reporters / total          # observed pro

```

```

denominator = 1 + z**2 / total          # Wilson denom
centre       = rate + z**2 / (2 * total) # shifted cent
half_width   = z * np.sqrt(rate*(1-rate)/total) # spread from
              + z**2 / (4*total**2))      # + small-sa
return rate, (centre - half_width)/denominator, (centre + half_width)

# ---- PED (performance-enhancing-drug) flag ----
STEROID_TERMS = ["trt", "testosterone", "hgh", "steroid", "anavar", "tren",
                 "nandrolone", "deca", "winstrol", "primo", "masteron", "mk-677", "mk677", "clen"]

def uses_ped(aid):
    """True if any confounder label for this author matches a PED/steroid
    profile = _by.get(aid) or {}
    for event in (profile.get("timeline_events") or []):
        if isinstance(event, dict):
            for conf in (event.get("confounding_variables") or []):
                if isinstance(conf, dict) and any(
                    term in (conf.get("specific_label") or "").lower(
                        return True
    return False

```

```

In [ ]: #####
# Cohort selection, injection route-confirmed filter, and author-level de
#####
_scaled = pd.read_json(scaled_output_path, lines=True)
_by = {r["author"]: parse_model_output(r["model_output"]) for _, r in _
_drug = dict(zip(_scaled["author"], _scaled["drug_group"]))

# --- administration by injection route-confirmed filter: drop explicit o
# injection route for oz/sema/weg ---
analysis_set_ids = [
    aid for aid in analysis_set_ids
    if "oral" not in str(_by.get(aid) or {}).get("route_of_administratio
    and (str(_by.get(aid) or {}).get("route_of_administration") or "").s
    or _drug.get(aid) == "reta")
]
print("route-confirmed per-group sizes:",
      dict(sorted(Counter(_drug.get(a) for a in analysis_set_ids).items())
print(f"route-confirmed total: {len(analysis_set_ids)}")

# --- demographic extractors - return raw values recorded. The only excep
# for DERIVED BMI, which keeps a minimal height>50, to limit noise/nonsen
def get_age(aid):
    v = ((_by.get(aid) or {}).get("patient_profile") or {}).get("age_year
    return float(v) if isinstance(v, (int, float)) else None
def get_sex(aid):
    s = ((_by.get(aid) or {}).get("patient_profile") or {}).get("sex_gen
    if s in ("m", "man", "male"): return "male"
    if s in ("f", "woman", "female"): return "female"
    return None
def get_height(aid):
    h = ((_by.get(aid) or {}).get("patient_profile") or {}).get("height_c
    return float(h) if isinstance(h, (int, float)) else None
def get_bmi(aid):
    """BMI if directly recorded, else DERIVED from height + first valid w
    Recorded BMI is returned as-is (type-checked only). For the DERIVED v
    require height>50cm - not a plausibility filter on real people, but a
    a garbage height doesn't blow up the kg/m^2 division into a nonsense
    prof = (_by.get(aid) or {}).get("patient_profile") or {}
    bmi = prof.get("bmi")

```

```

    if isinstance(bmi, (int, float)):
        return round(float(bmi), 1)
    height = get_height(aid)
    if height and height > 50:
        for e in ((_by.get(aid) or {}).get("timeline_events") or []):
            if isinstance(e, dict):
                weight = e.get("weight_kg")
                if weight is None and e.get("weight_lb") is not None:
                    weight = float(e["weight_lb"]) * 0.453592 # convert
                if isinstance(weight, (int, float)):
                    return round(weight / ((height/100) ** 2), 1) # BMI
        return None
def get_race(aid):
    r = ((_by.get(aid) or {}).get("patient_profile") or {}).get("race_eth")
    if isinstance(r, str) and r.strip() and r.strip().lower() not in ("no
        return r.strip()
    return None

age_map = {a: get_age(a) for a in analysis_set_ids}
sex_map = {a: get_sex(a) for a in analysis_set_ids}
height_map = {a: get_height(a) for a in analysis_set_ids}
bmi_map = {a: get_bmi(a) for a in analysis_set_ids}
race_map = {a: get_race(a) for a in analysis_set_ids}

# --- long-format effects table for the FULL injection confirmed cohort (
# with RAW adverse and desired effect labels. ---
rows = []
for aid in analysis_set_ids:
    p = _by.get(aid) or {}; drug = _drug.get(aid)
    for e in p.get("timeline_events") or []:
        if not isinstance(e, dict):
            continue
        for event, field in [("desired", "desired_effects"), ("adverse",
            for item in e.get(field) or []:
                if not isinstance(item, dict):
                    continue
                rows.append({
                    "author_id": aid, "drug_group": drug, "event": event,
                    "visualisation_group": (item.get("visualisation_group")
                    "specific_label": item.get("specific_label"),
                    "severity": item.get("severity"), "date": e.get("date")
                    "age": age_map.get(aid), "sex": sex_map.get(aid),
                })
effects_long = pd.DataFrame(rows)
print(f"effect mentions: {len(effects_long):,} "
      f"(desired {sum(effects_long['event']=='desired'):,} / adverse {sum

```

route-confirmed per-group sizes: {'oz': 403, 'reta': 392, 'sema': 342, 'we
g': 369}

route-confirmed total: 1506

effect mentions: 36,555 (desired 19,092 / adverse 17,463)

```

In [ ]: #####
# Explore recovery + completeness for each demographic field #
# to drive the cohort selection decision. #
#####
N = len(analysis_set_ids)
SATURATION_FLOOR = 30 #minimum authors required for cohort selection

FIELDS = {"age": age_map, "sex": sex_map, "height": height_map,

```

```

        "BMI": bmi_map, "race/ethnicity": race_map}
print(f"=== demographic recovery (route-confirmed, n={N}) ===")
print(f"{'field':16s} {'n':>5s} {'%':>6s}    per drug (oz/reta/sema/weg)")
for name, m in FIELDS.items():
    have      = {a for a in analysis_set_ids if m.get(a) is not None}
    by_drug   = [sum(1 for a in have if _drug.get(a) == dg) for dg in DRUG_]
    print(f"{'name':16s} {len(have):>5d} {100*len(have)/N:>5.1f}%    {by_drug}")

def cohort_with(*maps):
    """Authors having a non-None value in EVERY listed demographic map."""
    return {a for a in analysis_set_ids if all(m.get(a) is not None for m in maps)}

# candidate cohorts of increasing completeness
has_sex      = cohort_with(sex_map)
has_age      = cohort_with(age_map)
has_both     = cohort_with(sex_map, age_map)
has_sex_age_bmi = cohort_with(sex_map, age_map, bmi_map)
has_sex_age_height = cohort_with(sex_map, age_map, height_map)
full_demo    = cohort_with(sex_map, age_map, height_map, bmi_map)
# (race excluded from every cohort – recovered for almost no one, would c

print("\n=== cohort options (per-drug counts) ===")
cohort_options = [
    ("sex",          has_sex),          ("age",          has_a
    ("sex+age",      has_both),          ("sex+age+BMI",    has_s
    ("sex+age+height", has_sex_age_height), ("sex+age+ht+BMI", full_
]
for label, s in cohort_options:
    by_drug = dict(sorted(Counter(_drug.get(a) for a in s).items()))
    weg_ok  = "OK" if by_drug.get("weg", 0) >= SATURATION_FLOOR else f"<
    print(f" {label:16s} n={len(s):4d}    {by_drug}    weg: {weg_ok}")

# --- distributions within the chosen cohort ---
print(f"\n=== distributions within sex+age (n={len(has_both)}) ===")
for name, m in [("age", age_map), ("height", height_map), ("BMI", bmi_map)
    vals = [m[a] for a in has_both if m.get(a) is not None]
    if vals:
        print(f" {name:8s} n={len(vals):3d} median {np.median(vals):.1f}
              f"IQR {np.percentile(vals,25):.1f}–{np.percentile(vals,75):
              f"range {min(vals):.1f}–{max(vals):.1f}")
print(" sex:", dict(Counter(sex_map[a] for a in has_both if sex_map.get(
race_vals = [race_map[a] for a in has_both if race_map.get(a)]
print(f" race/ethnicity: {len(race_vals)} recorded",
      dict(Counter(race_vals)) if race_vals else "(none – generalisabilit

```

```

=== demographic recovery (route-confirmed, n=1506) ===
field          n      %    per drug (oz/reta/sema/weg)
age            450  29.9%  [143, 116, 63, 128]
sex            485  32.2%  [140, 127, 72, 146]
height         438  29.1%  [134, 127, 57, 120]
BMI            407  27.0%  [127, 118, 50, 112]
race/ethnicity    5   0.3%  [1, 2, 0, 2]

=== cohort options (per-drug counts) ===
sex            n= 485   {'oz': 146, 'reta': 140, 'sema': 127, 'weg': 7
2}   weg: OK
age            n= 450   {'oz': 128, 'reta': 143, 'sema': 116, 'weg': 6
3}   weg: OK
sex+age        n= 339   {'oz': 101, 'reta': 104, 'sema': 88, 'weg': 4
6}   weg: OK
sex+age+BMI    n= 212   {'oz': 62, 'reta': 67, 'sema': 59, 'weg': 24}
weg: < 30 FLOOR
sex+age+height n= 226   {'oz': 66, 'reta': 71, 'sema': 63, 'weg': 26}
weg: < 30 FLOOR
sex+age+ht+BMI n= 212   {'oz': 62, 'reta': 67, 'sema': 59, 'weg': 24}
weg: < 30 FLOOR

=== distributions within sex+age (n=339) ===
age            n=339  median 42.0  IQR 32.0–52.0  range 15.0–76.0
height         n=226  median 170.2  IQR 162.6–180.3  range 147.3–198.1
BMI            n=212  median 30.4  IQR 26.6–35.1  range 18.6–63.1
sex: {'female': 222, 'male': 117}
race/ethnicity: 2 recorded {'black': 1, 'white': 1}

```

```

In [ ]: #####
# Inspect the raw adverse and desired effect labels.#
# Define & apply the canonicalised events          #
#####

# distinct raw authors reporting a RAW adverse and desired effects
def raw_authors_reporting(effect, event, ids):
    ids = set(ids)
    return effects_long[(effects_long["event"] == event) &
                        (effects_long["visualisation_group"] == effect) &
                        (effects_long["author_id"].isin(ids))]["author_id"].nunique()

# --- list the raw labels + their author-level prevalence ---
print("=== RAW effect labels (sex+age), author-level prevalence – pre-can
for event in ["desired", "adverse"]:
    pool = effects_long[(effects_long["event"] == event) &
                        (effects_long["author_id"].isin(has_both))]
    counts = pool.groupby("visualisation_group")["author_id"].nunique().s
    print(f"\n {event.upper()} – {len(counts)} distinct raw labels:")
    for label, k in counts.items():
        print(f"    {label:38s} {k:4d}    ({100*k/len(has_both):.1f}%)"

# --- define and APPLY the canonical map ---
DESIRED_CANONICAL = {
    "improved satiety":      "reduced appetite",
    "portion control":      "reduced appetite",
    "psychological wellbeing": "improved quality of life",
    "improved mood":        "improved quality of life",
    "improved confidence":  "improved quality of life",
}
ADVERSE_CANONICAL = {

```

```

    "loss of appetite": "reduced appetite",
}
def canonical_label(label, event):
    """Map a raw effect label to its canonical form, per event type."""
    key = (label or "").lower().strip()
    if event == "desired": return DESIRED_CANONICAL.get(key, key)
    if event == "adverse": return ADVERSE_CANONICAL.get(key, key)
    return key

effects_long["visualisation_group_canonical"] = effects_long.apply(
    lambda r: canonical_label(r["visualisation_group"], r["event"]), axis=1
)
effects_long.to_csv(f"{gpt_result_dir}/effects_long_analysis_set.csv", index=False)

# --- verify exactly which labels changed ---
print("\n=== canonicalised labels that changed ===")
for event in ["desired", "adverse"]:
    changed = effects_long[(effects_long["event"] == event) &
                           (effects_long["visualisation_group"] != effects_long["visualisation_group_canonical"])]
    print(f" {event.upper()}:")
    print(changed.groupby(["visualisation_group", "visualisation_group_canonical"]).size())

# --- canonicalised reporting ---
def authors_reporting(effect, event, ids):
    """Distinct authors in `ids` reporting this CANONICAL effect for this event"""
    ids = set(ids)
    return effects_long[(effects_long["event"] == event) &
                        (effects_long["visualisation_group_canonical"] == effect) &
                        (effects_long["author_id"].isin(ids))]["author_id"].nunique()

def prevalence(effect, event, ids):
    """Fraction of a group who reported this canonical effect at least once"""
    return authors_reporting(effect, event, ids) / len(ids) if ids else 0

```


=== RAW effect labels (sex+age), author-level prevalence – pre-canonicalisation ===

DESIRED – 110 distinct raw labels:

weight loss	314	(92.6%)
reduced appetite	206	(60.8%)
reduced food noise	170	(50.1%)
improved quality of life	150	(44.2%)
dietary improvement	118	(34.8%)
portion control	101	(29.8%)
improved impulse control	93	(27.4%)
relieved gastrointestinal symptoms	89	(26.3%)
improved satiety	82	(24.2%)
weight maintenance	79	(23.3%)
psychological wellbeing	76	(22.4%)
increased physical activity	68	(20.1%)
improved blood glucose control	63	(18.6%)
reduced inflammation	55	(16.2%)
improved mood	47	(13.9%)
maintained energy	45	(13.3%)
improved confidence	40	(11.8%)
improved fatigue	37	(10.9%)
improved blood pressure	35	(10.3%)
reduced cholesterol	22	(6.5%)
improved sleep	14	(4.1%)
improved relationship with food	13	(3.8%)
reduced alcohol cravings	11	(3.2%)
reduced alcohol use	10	(2.9%)
improved diabetes symptoms	9	(2.7%)
menstrual changes	8	(2.4%)
sleep disturbance	7	(2.1%)
other	7	(2.1%)
improved sleep apnea	6	(1.8%)
improved lab results	6	(1.8%)
improved menstrual regularity	5	(1.5%)
reduced alcohol desire	5	(1.5%)
muscle loss	4	(1.2%)
improved sleep quality	4	(1.2%)
reduced alcohol craving	4	(1.2%)
improved menstrual symptoms	4	(1.2%)
improved resting heart rate	3	(0.9%)
improved pcos symptoms	3	(0.9%)
increased libido	3	(0.9%)
improved mobility	3	(0.9%)
weight plateau	3	(0.9%)
treatment continuation	3	(0.9%)
improved fatty liver	2	(0.6%)
reduced musculoskeletal pain	2	(0.6%)
craving changes	2	(0.6%)
improved acne	2	(0.6%)
loss of appetite	2	(0.6%)
improved sexual function	2	(0.6%)
improved liver enzymes	2	(0.6%)
improved iron status	2	(0.6%)
improved liver function	2	(0.6%)
muscle preservation	2	(0.6%)
improved libido	2	(0.6%)
reduced infections	1	(0.3%)
preserved muscle mass	1	(0.3%)
no appetite suppression	1	(0.3%)

reduced fluid retention	1	(0.3%)
transplant eligibility	1	(0.3%)
stable vision	1	(0.3%)
pcos	1	(0.3%)
smoking cessation	1	(0.3%)
relieved dermatologic symptoms	1	(0.3%)
reduced medication use	1	(0.3%)
reduced water retention	1	(0.3%)
reduced triglycerides	1	(0.3%)
reduced substance use	1	(0.3%)
reduced cravings	1	(0.3%)
reduced smoking	1	(0.3%)
reduced hair loss	1	(0.3%)
reduced resting heart rate	1	(0.3%)
reduced night sweats	1	(0.3%)
reduced edema	1	(0.3%)
body recomposition	1	(0.3%)
muscle gain	1	(0.3%)
improved hair growth	1	(0.3%)
improved focus	1	(0.3%)
improved fertility	1	(0.3%)
improved fatty liver disease	1	(0.3%)
improved colorectal health	1	(0.3%)
improved colon health	1	(0.3%)
improved cognitive clarity	1	(0.3%)
improved cognition	1	(0.3%)
improved cardiovascular health	1	(0.3%)
improved cardiometabolic markers	1	(0.3%)
improved breathing	1	(0.3%)
improved body composition	1	(0.3%)
improved adhd symptoms	1	(0.3%)
headaches	1	(0.3%)
hair regrowth	1	(0.3%)
dysgeusia	1	(0.3%)
improved gout symptoms	1	(0.3%)
improved headaches	1	(0.3%)
maintained strength	1	(0.3%)
improved insulin resistance	1	(0.3%)
increased lean mass	1	(0.3%)
improved vision	1	(0.3%)
improved urinary symptoms	1	(0.3%)
improved tolerability	1	(0.3%)
improved stroke symptoms	1	(0.3%)
body size reduction	1	(0.3%)
improved retinopathy	1	(0.3%)
improved respiratory symptoms	1	(0.3%)
improved plantar fasciitis	1	(0.3%)
improved neuropathy symptoms	1	(0.3%)
improved migraine symptoms	1	(0.3%)
improved menopause symptoms	1	(0.3%)
improved macular edema	1	(0.3%)
improved liver disease	1	(0.3%)
improved laboratory markers	1	(0.3%)
improved skin	1	(0.3%)

ADVERSE – 166 distinct raw labels:

gastrointestinal issues	248	(73.2%)
fatigue	144	(42.5%)
no appetite suppression	106	(31.3%)
loss of appetite	77	(22.7%)

weight gain	77	(22.7%)
dose intolerance	64	(18.9%)
headaches	60	(17.7%)
low mood	55	(16.2%)
weight plateau	54	(15.9%)
sleep disturbance	48	(14.2%)
dehydration	46	(13.6%)
dysgeusia	43	(12.7%)
muscle loss	41	(12.1%)
increased sugar appetite	39	(11.5%)
body dysmorphia	37	(10.9%)
skin sensitivity	37	(10.9%)
injection site inflammation	32	(9.4%)
reduced appetite	30	(8.8%)
elevated heart rate	30	(8.8%)
other	29	(8.6%)
cold temperature sensitivity	27	(8.0%)
menstrual changes	27	(8.0%)
treatment failure	21	(6.2%)
hair loss	21	(6.2%)
dizziness	19	(5.6%)
weight maintenance	14	(4.1%)
anxiety	12	(3.5%)
increased food noise	12	(3.5%)
loose skin	12	(3.5%)
hypoglycemia	11	(3.2%)
reduced food noise	11	(3.2%)
musculoskeletal pain	7	(2.1%)
hypoglycaemia	7	(2.1%)
increased appetite	6	(1.8%)
hypoglycemia symptoms	6	(1.8%)
reduced libido	6	(1.8%)
alcohol intolerance	6	(1.8%)
sexual dysfunction	6	(1.8%)
increased cravings	4	(1.2%)
vision changes	4	(1.2%)
hypotension	4	(1.2%)
increased calorie intake	4	(1.2%)
urinary frequency	3	(0.9%)
cognitive impairment	3	(0.9%)
heat intolerance	3	(0.9%)
lightheadedness	3	(0.9%)
fluid retention	3	(0.9%)
syncope	3	(0.9%)
anhedonia	3	(0.9%)
improved blood glucose control	3	(0.9%)
orthostatic hypotension	3	(0.9%)
worsened blood pressure	2	(0.6%)
worsened food noise	2	(0.6%)
worsened blood glucose control	2	(0.6%)
persistent food noise	2	(0.6%)
weight loss	2	(0.6%)
visual disturbance	2	(0.6%)
jaw pain	2	(0.6%)
urinary changes	2	(0.6%)
skin changes	2	(0.6%)
muscle discomfort	2	(0.6%)
relieved gastrointestinal symptoms	2	(0.6%)
reduced mobility	2	(0.6%)
acne	2	(0.6%)

irritability	2	(0.6%)
hypoglycaemia symptoms	2	(0.6%)
food aversion	2	(0.6%)
alcohol use	2	(0.6%)
back pain	2	(0.6%)
body aches	2	(0.6%)
cognitive symptoms	2	(0.6%)
drug interaction	2	(0.6%)
excess skin	2	(0.6%)
flushing	2	(0.6%)
worsened pain	2	(0.6%)
increased food reward	1	(0.3%)
smell sensitivity	1	(0.3%)
compulsive shopping	1	(0.3%)
constipation	1	(0.3%)
skin laxity	1	(0.3%)
decreased libido	1	(0.3%)
shingles	1	(0.3%)
device or administration issue	1	(0.3%)
sensory change	1	(0.3%)
sedation	1	(0.3%)
salivary gland issues	1	(0.3%)
rhinitis	1	(0.3%)
dietary improvement	1	(0.3%)
reduced motivation	1	(0.3%)
disordered eating	1	(0.3%)
hypertension	1	(0.3%)
reduced inflammation	1	(0.3%)
dry mouth	1	(0.3%)
reduced enjoyment of food	1	(0.3%)
reduced efficacy	1	(0.3%)
electrolyte imbalance	1	(0.3%)
sleep-disordered breathing	1	(0.3%)
stimulant sensitivity	1	(0.3%)
raynaud phenomenon	1	(0.3%)
increased alcohol sensitivity	1	(0.3%)
worsened inflammation	1	(0.3%)
alcohol tolerance reduction	1	(0.3%)
worsened cholesterol	1	(0.3%)
worsened breathing	1	(0.3%)
improved impulse control	1	(0.3%)
altered alcohol response	1	(0.3%)
worsened binge eating	1	(0.3%)
improved mood	1	(0.3%)
increased alcohol desire	1	(0.3%)
body composition change	1	(0.3%)
body composition changes	1	(0.3%)
water retention	1	(0.3%)
body odor	1	(0.3%)
brain fog	1	(0.3%)
urticaria	1	(0.3%)
urine color change	1	(0.3%)
breast imaging changes	1	(0.3%)
chills	1	(0.3%)
cognitive issues	1	(0.3%)
reduced alcohol effect	1	(0.3%)
psychological wellbeing	1	(0.3%)
increased impulsive spending	1	(0.3%)
flu-like symptoms	1	(0.3%)
low libido	1	(0.3%)

heart rate variability change	1	(0.3%)
food noise recurrence	1	(0.3%)
gallbladder disease	1	(0.3%)
leukopenia	1	(0.3%)
laboratory abnormality	1	(0.3%)
lab abnormality	1	(0.3%)
kidney pain	1	(0.3%)
kidney concerns	1	(0.3%)
joint pain	1	(0.3%)
jitteriness	1	(0.3%)
gout	1	(0.3%)
alcohol effects change	1	(0.3%)
hair growth changes	1	(0.3%)
injection discomfort	1	(0.3%)
halitosis	1	(0.3%)
increased substance cravings	1	(0.3%)
increased intraocular pressure	1	(0.3%)
increased inflammation	1	(0.3%)
heart rate variability changes	1	(0.3%)
muscle cramps	1	(0.3%)
pruritus	1	(0.3%)
falls	1	(0.3%)
pots	1	(0.3%)
elevated intraocular pressure	1	(0.3%)
palpitations	1	(0.3%)
improved blood pressure	1	(0.3%)
elevated liver enzymes	1	(0.3%)
high blood pressure	1	(0.3%)
olfactory sensitivity	1	(0.3%)
olfactory changes	1	(0.3%)
no food noise reduction	1	(0.3%)
no food noise control	1	(0.3%)
excessive weight loss	1	(0.3%)
neutropenia	1	(0.3%)
neuropathy	1	(0.3%)
neurological symptoms	1	(0.3%)
nasal congestion	1	(0.3%)
facial appearance changes	1	(0.3%)
muscle twitching	1	(0.3%)
muscle soreness	1	(0.3%)
fainting	1	(0.3%)
pain	1	(0.3%)

=== canonicalisedlabels that changed ===

DESIRED:

visualisation_group	visualisation_group_canonical	
improved confidence	improved quality of life	145
improved mood	improved quality of life	212
improved satiety	reduced appetite	479
portion control	reduced appetite	633
psychological wellbeing	improved quality of life	314

ADVERSE:

visualisation_group	visualisation_group_canonical	
loss of appetite	reduced appetite	526

```
In [ ]: #####
# Checking reta's canonicalised effects by prevalence.#
#####
reta_ids = [a for a in has_both if _drug.get(a) == "reta"]
print(f"retatrutide authors (sex+age): {len(reta_ids)}\n")
```

```

def top_reta_effects(event, n=11):
    """The n most prevalent canonical effects among reta authors (sex+age
    n=11 rather than 10 avoids splitting a tied point estimate at the 10th
    position)"""
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_ids))]
    prevs = [(eff, prevalence(eff, event, reta_ids))
              for eff in reta_pool["visualisation_group_canonical"].unique()]
    prevs.sort(key=lambda pair: pair[1], reverse=True)  # most prevalent
    return [eff for eff, _ in prevs[:n]]

for event in ["adverse", "desired"]:
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_ids))]
    effects = reta_pool["visualisation_group_canonical"].value_counts().h
    rows = []
    for eff in effects:
        reporters = authors_reporting(eff, event, reta_ids)
        rate, ci_low, ci_high = wilson(reporters, len(reta_ids))
        rows.append(dict(effect=eff, reporters=reporters, rate=rate, lo=ci_low, hi=ci_high))
    df = pd.DataFrame(rows).sort_values("rate", ascending=False)
    df["prevalence"] = df.apply(
        lambda r: f"{r.rate*100:.1f}% [{r.lo*100:.0f}-{r.hi*100:.0f}]", axis=1
    )
    print(f"=== RETATRUTIDE {event.upper()} by prevalence (sex+age, canonical effects)")
    print(df[["effect", "prevalence"]].to_string(index=False), "\n")

```

retatrutide authors (sex+age): 104

=== RETATRUTIDE ADVERSE by prevalence (sex+age, canonical) ===

	effect	prevalence
gastrointestinal issues	58.7% [49–68]	(n=61)
fatigue	46.2% [37–56]	(n=48)
no appetite suppression	30.8% [23–40]	(n=32)
reduced appetite	27.9% [20–37]	(n=29)
dose intolerance	24.0% [17–33]	(n=25)
elevated heart rate	22.1% [15–31]	(n=23)
sleep disturbance	21.2% [14–30]	(n=22)
skin sensitivity	21.2% [14–30]	(n=22)
weight gain	18.3% [12–27]	(n=19)
low mood	17.3% [11–26]	(n=18)
muscle loss	17.3% [11–26]	(n=18)
dehydration	15.4% [10–24]	(n=16)
headaches	10.6% [6–18]	(n=11)
weight plateau	10.6% [6–18]	(n=11)
injection site inflammation	10.6% [6–18]	(n=11)
cold temperature sensitivity	9.6% [5–17]	(n=10)
increased sugar appetite	4.8% [2–11]	(n=5)
sexual dysfunction	4.8% [2–11]	(n=5)
treatment failure	3.8% [2–9]	(n=4)
hypoglycemia	3.8% [2–9]	(n=4)

=== RETATRUTIDE DESIRED by prevalence (sex+age, canonical) ===

	effect	prevalence
weight loss	88.5% [81–93]	(n=92)
reduced appetite	71.2% [62–79]	(n=74)
reduced food noise	53.8% [44–63]	(n=56)
improved quality of life	51.9% [42–61]	(n=54)
improved impulse control	26.9% [19–36]	(n=28)
dietary improvement	26.9% [19–36]	(n=28)
maintained energy	21.2% [14–30]	(n=22)
relieved gastrointestinal symptoms	21.2% [14–30]	(n=22)
weight maintenance	20.2% [14–29]	(n=21)
improved blood glucose control	14.4% [9–22]	(n=15)
reduced inflammation	14.4% [9–22]	(n=15)
increased physical activity	13.5% [8–21]	(n=14)
improved fatigue	11.5% [7–19]	(n=12)
improved blood pressure	10.6% [6–18]	(n=11)
reduced cholesterol	8.7% [5–16]	(n=9)
reduced alcohol cravings	3.8% [2–9]	(n=4)
sleep disturbance	3.8% [2–9]	(n=4)
improved sleep	3.8% [2–9]	(n=4)
improved liver function	1.9% [1–7]	(n=2)
craving changes	1.0% [0–5]	(n=1)

```
In [ ]: #####
# Splits reta and pooled sema-family by PED use. #
#####
reta_all  = [a for a in has_both if _drug.get(a) == "reta"]
sema_all  = [a for a in has_both if _drug.get(a) in SEMA_FAMILY] # p
reta_ped  = [a for a in reta_all if uses_ped(a)]
reta_noped = [a for a in reta_all if not uses_ped(a)]
sema_ped  = [a for a in sema_all if uses_ped(a)]
sema_noped = [a for a in sema_all if not uses_ped(a)]
print("sex+age PED split:")
print(f"  reta: {len(reta_all)} = {len(reta_ped)} PED + {len(reta_noped)}")
```

```

print(f"  sema: {len(sema_all)} = {len(sema_ped)} PED + {len(sema_noped)}

def rate_str(effect, event, ids):
    """'rate [lo-hi]' percentage string for a subgroup, or '-' if empty."
    reporters = authors_reporting(effect, event, ids); total = len(ids)
    if total == 0:
        return "-"
    rate, lo, hi = wilson(reporters, total)
    return f"{rate*100:.1f} [{lo*100:.0f}-{hi*100:.0f}]"

def deconfound(event, top_n=20):
    """Four-subgroup prevalence table; p/q test the DECONFOUNDED comparis
    (non-PED reta vs non-PED sema), BH-corrected across the effects shown
    pool = effects_long[(effects_long["event"] == event) &
                        (effects_long["author_id"].isin(has_both))]
    effects = pool["visualisation_group_canonical"].value_counts().head(t
    rows = []
    for eff in effects:
        # reported vs not, non-PED reta vs non-PED sema
        reta_k = authors_reporting(eff, event, reta_noped); reta_n = len(
        sema_k = authors_reporting(eff, event, sema_noped); sema_n = len(
        _, pval = fisher_exact([[reta_k, reta_n - reta_k], [sema_k, sema_
        rows.append(dict(effect=eff,
                        reta_PED = rate_str(eff, event, reta_ped),
                        reta_nonPED= rate_str(eff, event, reta_noped),
                        sema_PED = rate_str(eff, event, sema_ped),
                        sema_nonPED= rate_str(eff, event, sema_noped),
                        p=pval))
    df = pd.DataFrame(rows)
    df["q_BH"] = multipletests(df["p"], method="fdr_bh")[1] #False disc
    df["sig"] = (df["q_BH"] < 0.05).astype(int)
    return df.sort_values("q_BH") #most signi

for event in ["adverse", "desired"]:
    df = deconfound(event)
    out = df.copy()
    out["p"] = out["p"].map(lambda v: f"{v:.3f}"); out["q(BH)"] = out["q_
    print(f"\n=== {event.upper()} - sex+age drug×PED (canonical; p/q = no
    print(out[["effect", "reta_PED", "reta_nonPED", "sema_PED", "sema_non
    df.to_csv(f"{gpt_result_dir}/hasboth_PEDdeconfound_canonical_{event}.
```


sex+age PED split:

reta: 104 = 38 PED + 66 non-PED

sema: 235 = 32 PED + 203 non-PED

=== ADVERSE – sex+age drug×PED (canonical; p/q = non-PED reta vs non-PED sema) ===

ED	p	q(BH)	sig	effect	reta_PED	reta_nonPED	sema_PED	sema_nonPED
6]	0.000	0.000	1	elevated heart rate	21.1 [11–36]	22.7 [14–34]	3.1 [1–16]	3.0 [1–6]
3]	0.001	0.009	1	gastrointestinal issues	63.2 [47–77]	56.1 [44–67]	93.8 [80–98]	77.3 [71–83]
0]	0.001	0.009	1	skin sensitivity	21.1 [11–36]	21.2 [13–33]	9.4 [3–24]	5.9 [3–10]
9]	0.021	0.104	0	dysgeusia	10.5 [4–24]	3.0 [1–10]	31.2 [18–49]	13.3 [9–19]
6]	0.059	0.235	0	headaches	13.2 [6–27]	9.1 [4–18]	28.1 [16–45]	19.7 [15–26]
8]	0.107	0.355	0	weight gain	28.9 [17–45]	12.1 [6–22]	43.8 [28–61]	21.7 [17–28]
6]	0.149	0.425	0	body dysmorphia	5.3 [1–17]	4.5 [2–13]	31.2 [18–49]	10.8 [7–16]
8]	0.177	0.442	0	increased sugar appetite	2.6 [0–13]	6.1 [2–15]	25.0 [13–42]	12.8 [9–18]
3]	0.243	0.503	0	weight plateau	10.5 [4–24]	10.6 [5–20]	25.0 [13–42]	17.2 [13–23]
1]	0.260	0.503	0	dose intolerance	28.9 [17–45]	21.2 [13–33]	25.0 [13–42]	15.3 [11–21]
5]	0.277	0.503	0	sleep disturbance	31.6 [19–47]	15.2 [8–26]	15.6 [7–32]	10.3 [7–15]
4]	0.303	0.505	0	menstrual changes	2.6 [0–13]	4.5 [2–13]	12.5 [5–28]	9.4 [6–14]
1]	0.373	0.574	0	treatment failure	5.3 [1–17]	3.0 [1–10]	12.5 [5–28]	6.4 [4–11]
2]	0.575	0.639	0	muscle loss	39.5 [26–55]	4.5 [2–13]	25.0 [13–42]	7.4 [5–12]
2]	0.573	0.639	0	low mood	13.2 [6–27]	19.7 [12–31]	12.5 [5–28]	16.3 [12–22]
8]	0.515	0.639	0	dehydration	26.3 [15–42]	9.1 [4–18]	12.5 [5–28]	12.8 [9–18]
4]	0.522	0.639	0	reduced appetite	36.8 [23–53]	22.7 [14–34]	37.5 [23–55]	27.1 [21–34]
4]	0.572	0.639	0	orthostatic hypotension	0.0 [0–9]	1.5 [0–8]	0.0 [0–11]	1.0 [0–4]
7]	0.775	0.816	0	fatigue	52.6 [37–68]	42.4 [31–54]	43.8 [28–61]	40.4 [34–47]
6]	1.000	1.000	0	no appetite suppression	34.2 [21–50]	28.8 [19–41]	46.9 [31–64]	29.1 [23–36]

=== DESIRED – sex+age drug×PED (canonical; p/q = non-PED reta vs non-PED sema) ===

sema_nonPED	p	q(BH)	sig	effect	reta_PED	reta_nonPED	sema_PED
94.1 [90–97]	0.035	0.693	0	weight loss	94.7 [83–99]	84.8 [74–92]	96.9 [84–99]
4.4 [2–8]	0.459	0.807	0	reduced alcohol use	0.0 [0–9]	1.5 [0–8]	0.0 [0–11]
10.8 [7–16]	0.637	0.807	0	improved fatigue	18.4 [9–33]	7.6 [3–17]	9.4 [3–24]
				reduced cholesterol	10.5 [4–24]	7.6 [3–17]	6.2 [2–20]

5.4 [3-9]	0.551	0.807	0			
	improved blood pressure			7.9 [3-21]	12.1 [6-22]	12.5 [5-28]
9.9 [6-15]	0.644	0.807	0			
	maintained energy			28.9 [17-45]	16.7 [10-27]	6.2 [2-20]
10.3 [7-15]	0.190	0.807	0			
	increased physical activity			18.4 [9-33]	10.6 [5-20]	43.8 [28-61]
19.7 [15-26]	0.097	0.807	0			
	improved sleep			7.9 [3-21]	1.5 [0-8]	6.2 [2-20]
3.9 [2-8]	0.461	0.807	0			
	improved diabetes symptoms			0.0 [0-9]	0.0 [0-6]	3.1 [1-16]
3.9 [2-8]	0.206	0.807	0			
	improved impulse control			39.5 [26-55]	19.7 [12-31]	43.8 [28-61]
25.1 [20-32]	0.409	0.807	0			
	dietary improvement			28.9 [17-45]	25.8 [17-37]	50.0 [34-66]
36.5 [30-43]	0.134	0.807	0			
	relieved gastrointestinal symptoms			18.4 [9-33]	22.7 [14-34]	34.4 [20-52]
27.6 [22-34]	0.521	0.807	0			
	reduced food noise			55.3 [40-70]	53.0 [41-65]	59.4 [42-74]
46.8 [40-54]	0.398	0.807	0			
	improved quality of life			57.9 [42-72]	48.5 [37-60]	65.6 [48-80]
55.7 [49-62]	0.323	0.807	0			
	reduced appetite			68.4 [53-81]	72.7 [61-82]	87.5 [72-95]
69.5 [63-75]	0.646	0.807	0			
	weight maintenance			28.9 [17-45]	15.2 [8-26]	43.8 [28-61]
21.7 [17-28]	0.292	0.807	0			
	improved blood glucose control			10.5 [4-24]	16.7 [10-27]	25.0 [13-42]
19.7 [15-26]	0.718	0.845	0			
	reduced alcohol cravings			5.3 [1-17]	3.0 [1-10]	3.1 [1-16]
3.0 [1-6]	1.000	1.000	0			
	improved relationship with food			0.0 [0-9]	3.0 [1-10]	6.2 [2-20]
4.4 [2-8]	1.000	1.000	0			
	reduced inflammation			13.2 [6-27]	15.2 [8-26]	31.2 [18-49]
14.8 [11-20]	1.000	1.000	0			

```
In [ ]: #####
# Reta PED vs non-PED #
#####
print(f"retatrutide (sex+age): {len(reta_ped)} PED vs {len(reta_noped)} non-PED")
ped_driven = {}
for event in ["adverse", "desired"]:
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_all))]
    effects = reta_pool["visualisation_group_canonical"].value_counts().h
    rows = []
    for eff in effects:
        ped_k = authors_reporting(eff, event, reta_ped); ped_rate,
        noped_k = authors_reporting(eff, event, reta_noped); noped_rate,
        _, pval = fisher_exact([[ped_k, len(reta_ped)-ped_k], [noped_k, l
        rows.append(dict(effect=eff, PED=ped_rate, PED_lo=plo, PED_hi=phi
                           nonPED=noped_rate, nonPED_lo=nlo, nonPED_hi=nhi,
                           direction="higher in PED" if ped_rate > noped_ra
                           p=pval))
    df = pd.DataFrame(rows)
    df["q_BH"] = multipletests(df["p"], method="fdr_bh")[1]
    df["sig"] = (df["q_BH"] < 0.05).astype(int)
    df = df.sort_values("q_BH")
    out = df.copy()
    out["PED %"] = out.apply(lambda r: f"{r.PED*100:.1f}" [{r.PED_lo*1
    out["non-PED %"] = out.apply(lambda r: f"{r.nonPED*100:.1f}" [{r.nonPE
    out["p"] = out["p"].map(lambda v: f"{v:.3f}"); out["q(BH)"] = out["q_
```

```
print(f"=== {event.upper()} – reta PED vs non-PED (sex+age, canonical  
print(out[["effect", "PED %", "non-PED %", "direction", "p", "q(BH)",  
ped_driven[event] = df[df["sig"] == 1]["effect"].tolist()  
print("Effects differing by PED status WITHIN reta (BH q<0.05) = candidat  
print(f"  adverse: {ped_driven['adverse']}\n  desired: {ped_driven['desir
```

retatrutide (sex+age): 38 PED vs 66 non-PED

=== ADVERSE – reta PED vs non-PED (sex+age, canonical, sorted by q) ===					
			effect	PED %	non-PED % direction
p	q(BH)	sig			
0.000	0.000	1	muscle loss	39.5 [26–55]	4.5 [2–13] higher in PED
0.016	0.201	0	reduced libido	10.5 [4–24]	0.0 [0–6] higher in PED
0.025	0.210	0	dehydration	26.3 [15–42]	9.1 [4–18] higher in PED
0.039	0.242	0	weight gain	28.9 [17–45]	12.1 [6–22] higher in PED
0.058	0.292	0	sexual dysfunction	10.5 [4–24]	1.5 [0–8] higher in PED
0.079	0.330	0	sleep disturbance	31.6 [19–47]	15.2 [8–26] higher in PED
0.165	0.479	0	cold temperature sensitivity	15.8 [7–30]	6.1 [2–15] higher in PED
0.173	0.479	0	reduced appetite	36.8 [23–53]	22.7 [14–34] higher in PED
0.137	0.479	0	increased food noise	7.9 [3–21]	1.5 [0–8] higher in PED
0.204	0.510	0	injection site inflammation	15.8 [7–30]	7.6 [3–17] higher in PED
0.475	0.869	0	dose intolerance	28.9 [17–45]	21.2 [13–33] higher in PED
0.436	0.869	0	low mood	13.2 [6–27]	19.7 [12–31] higher in non-PED
0.622	0.869	0	treatment failure	5.3 [1–17]	3.0 [1–10] higher in PED
0.660	0.869	0	no appetite suppression	34.2 [21–50]	28.8 [19–41] higher in PED
0.525	0.869	0	headaches	13.2 [6–27]	9.1 [4–18] higher in PED
0.414	0.869	0	fatigue	52.6 [37–68]	42.4 [31–54] higher in PED
0.622	0.869	0	hypoglycemia	5.3 [1–17]	3.0 [1–10] higher in PED
0.650	0.869	0	increased sugar appetite	2.6 [0–13]	6.1 [2–15] higher in non-PED
0.539	0.869	0	gastrointestinal issues	63.2 [47–77]	56.1 [44–67] higher in PED
1.000	1.000	0	elevated heart rate	21.1 [11–36]	22.7 [14–34] higher in non-PED
1.000	1.000	0	other	7.9 [3–21]	7.6 [3–17] higher in PED
1.000	1.000	0	weight plateau	10.5 [4–24]	10.6 [5–20] higher in non-PED
1.000	1.000	0	worsened blood glucose control	0.0 [0–9]	1.5 [0–8] higher in non-PED
1.000	1.000	0	skin sensitivity	21.1 [11–36]	21.2 [13–33] higher in non-PED
1.000	1.000	0	anhedonia	2.6 [0–13]	1.5 [0–8] higher in PED

=== DESIRED – reta PED vs non-PED (sex+age, canonical, sorted by q) ===					
			effect	PED %	non-PED % direc
tion	p	q(BH)	sig		
			improved sleep	7.9 [3–21]	1.5 [0–8] higher in

PED 0.137	0.686	0	improved impulse control	39.5 [26–55]	19.7 [12–31]	higher in
PED 0.039	0.686	0	sleep disturbance	7.9 [3–21]	1.5 [0–8]	higher in
PED 0.137	0.686	0	improved fatigue	18.4 [9–33]	7.6 [3–17]	higher in
PED 0.117	0.686	0	weight maintenance	28.9 [17–45]	15.2 [8–26]	higher in
PED 0.128	0.686	0	weight loss	94.7 [83–99]	84.8 [74–92]	higher in
PED 0.203	0.756	0	maintained energy	28.9 [17–45]	16.7 [10–27]	higher in
PED 0.212	0.756	0	improved liver enzymes	2.6 [0–13]	0.0 [0–6]	higher in
PED 0.365	0.843	0	craving changes	2.6 [0–13]	0.0 [0–6]	higher in
PED 0.365	0.843	0	increased physical activity	18.4 [9–33]	10.6 [5–20]	higher in
PED 0.371	0.843	0	other	0.0 [0–9]	4.5 [2–13]	higher in non
–PED 0.298	0.843	0	improved quality of life	57.9 [42–72]	48.5 [37–60]	higher in
PED 0.417	0.869	0	reduced cholesterol	10.5 [4–24]	7.6 [3–17]	higher in
PED 0.721	1.000	0	improved sleep quality	2.6 [0–13]	1.5 [0–8]	higher in
PED 1.000	1.000	0	improved blood pressure	7.9 [3–21]	12.1 [6–22]	higher in non
–PED 0.742	1.000	0	relieved gastrointestinal symptoms	18.4 [9–33]	22.7 [14–34]	higher in non
–PED 0.804	1.000	0	reduced alcohol cravings	5.3 [1–17]	3.0 [1–10]	higher in
PED 0.622	1.000	0	improved liver function	2.6 [0–13]	1.5 [0–8]	higher in
PED 1.000	1.000	0	dietary improvement	28.9 [17–45]	25.8 [17–37]	higher in
PED 0.819	1.000	0	reduced food noise	55.3 [40–70]	53.0 [41–65]	higher in
PED 0.841	1.000	0	treatment continuation	2.6 [0–13]	1.5 [0–8]	higher in
PED 1.000	1.000	0	reduced appetite	68.4 [53–81]	72.7 [61–82]	higher in non
–PED 0.659	1.000	0	reduced cravings	0.0 [0–9]	1.5 [0–8]	higher in non
–PED 1.000	1.000	0	reduced inflammation	13.2 [6–27]	15.2 [8–26]	higher in non
–PED 1.000	1.000	0	improved blood glucose control	10.5 [4–24]	16.7 [10–27]	higher in non
–PED 0.564	1.000	0				

Effects differing by PED status WITHIN reta (BH $q < 0.05$) = candidate artefacts:

adverse: ['muscle loss']
desired: []

```
In [ ]: #####
# DEMOGRAPHICS DASHBOARD (visual summary of the recovery EDA + cohort cho
#####

dash_fields = ["age", "sex", "height", "BMI", "race/ethnicity"]
```

```

field_have = {f: {a for a in analysis_set_ids if FIELDS[f].get(a) is not None} for f in dash_fields}
field_pct = [100*len(field_have[f])/N for f in dash_fields]
field_n = [len(field_have[f]) for f in dash_fields]
per_drug = {f: [sum(1 for a in field_have[f] if _drug.get(a) == dg) for dg in DRUG_ORDER] for f in dash_fields}

dash_cohorts = ["injection\nroute-conf.", "sex", "age", "sex+age", "sex+age+route"]
dash_cohort_n = [N, len(has_sex), len(has_age), len(has_both), len(full_data)]

fig = plt.figure(figsize=(16, 10))
gs = GridSpec(2, 3, figure=fig, hspace=0.35, wspace=0.3)

# (a) recovery rate per field
ax1 = fig.add_subplot(gs[0, 0])
bars = ax1.barh(dash_fields, field_pct, color=["#4daf4a" if p > 25 else "#e41a1c"] for bar, n, pct in zip(bars, field_n, field_pct)):
    ax1.text(pct+0.25, bar.get_y()+bar.get_height()/2, f"{pct:.1f}% (n={n})")
ax1.set_xlim(0, 40); ax1.set_xlabel("% of cohort with field recorded"); ax1.set_title("Demographic recovery rate", fontweight="bold", fontsize=11)

# (b) cohort size by completeness
ax2 = fig.add_subplot(gs[0, 1])
bars = ax2.bar(range(len(dash_cohorts)), dash_cohort_n, color="#377eb8", for bar, n in zip(bars, dash_cohort_n):
    ax2.text(bar.get_x()+bar.get_width()/2, n+20, f"{n}", ha="center", fontweight="bold")
ax2.set_xticks(range(len(dash_cohorts))); ax2.set_xticklabels(dash_cohorts); ax2.set_ylabel("authors"); ax2.set_title("Cohort size by demographic completeness", fontweight="bold", fontsize=11)

# (c) recovery per drug group
ax3 = fig.add_subplot(gs[0, 2])
x = np.arange(len(dash_fields)); w = 0.2
for i, dg in enumerate(DRUG_ORDER):
    ax3.bar(x + (i-1.5)*w, [per_drug[f][i] for f in dash_fields], w, color=DRUG_COLOUR[dg], label=dg.upper(), alpha=0.85)
ax3.set_xticks(x); ax3.set_xticklabels(dash_fields, fontsize=8, rotation=45); ax3.set_ylabel("authors with field"); ax3.set_title("Recovery by drug group", fontweight="bold", fontsize=11)
ax3.legend(fontsize=8, ncol=2)

# (d) age distribution (sex+age)
ax4 = fig.add_subplot(gs[1, 0])
ages = [age_map[a] for a in has_both if age_map.get(a) is not None]
ax4.hist(ages, bins=15, color="#984ea3", alpha=0.75, edgecolor="white")
ax4.axvline(np.median(ages), color="black", ls="--", lw=1.5, label=f"median")
ax4.set_xlabel("age (years)"); ax4.set_ylabel("authors"); ax4.set_title("Age distribution (sex+age)", fontweight="bold", fontsize=11)

# (e) baseline BMI distribution (sex+age) with WHO category thresholds
ax5 = fig.add_subplot(gs[1, 1])
bmis = [bmi_map[a] for a in has_both if bmi_map.get(a) is not None]
ax5.hist(bmis, bins=15, color="#ff7f00", alpha=0.75, edgecolor="white")
for thr, lab in [(18.5, "underweight"), (25, "normal"), (30, "overweight")]:
    ax5.axvline(thr, color="grey", ls=":", lw=1)
    ax5.text(thr-0.4, ax5.get_ylim()[1]*0.9, lab, fontsize=7, rotation=90)
ax5.axvline(np.median(bmis), color="black", ls="--", lw=1.5, label=f"median")
ax5.set_xlabel("BMI"); ax5.set_ylabel("authors"); ax5.set_title("Baseline BMI distribution (sex+age)", fontweight="bold", fontsize=11)

# (f) sex composition per drug (sex+age)
ax6 = fig.add_subplot(gs[1, 2])
female = [sum(1 for a in has_both if _drug.get(a)==dg and sex_map.get(a)=='F') for dg in DRUG_ORDER]
male = [sum(1 for a in has_both if _drug.get(a)==dg and sex_map.get(a)=='M') for dg in DRUG_ORDER]

```

```

ax6.bar(DRUG_ORDER, female, color="#c2185b", alpha=0.8, label="female")
ax6.bar(DRUG_ORDER, male, bottom=female, color="#1565c0", alpha=0.8, label="male")
for i, dg in enumerate(DRUG_ORDER):
    tot = female[i]+male[i]
    if tot: ax6.text(i, tot+1, f"{100*male[i]/tot:.0f}% M", ha="center",
ax6.set_ylabel("authors"); ax6.set_xticklabels([d.upper() for d in DRUG_ORDER])
ax6.set_title("Sex composition by drug (sex+age)", fontweight="bold", fontstyle="italic")

for label, ax in zip("abcdef", [ax1, ax2, ax3, ax4, ax5, ax6]):
    ax.text(-0.08, 1.08, f"({label})", transform=ax.transAxes,
    fontsize=11, fontweight="bold", va="top", ha="right")
fig.suptitle("GLP-1 Reddit cohort – demographic recovery & composition",
    fontsize=15, fontweight="bold", y=0.98)
plt.savefig(f"{gpt_result_dir}/demographics_dashboard.png", dpi=140, bbox_inches="tight")
plt.show()

```

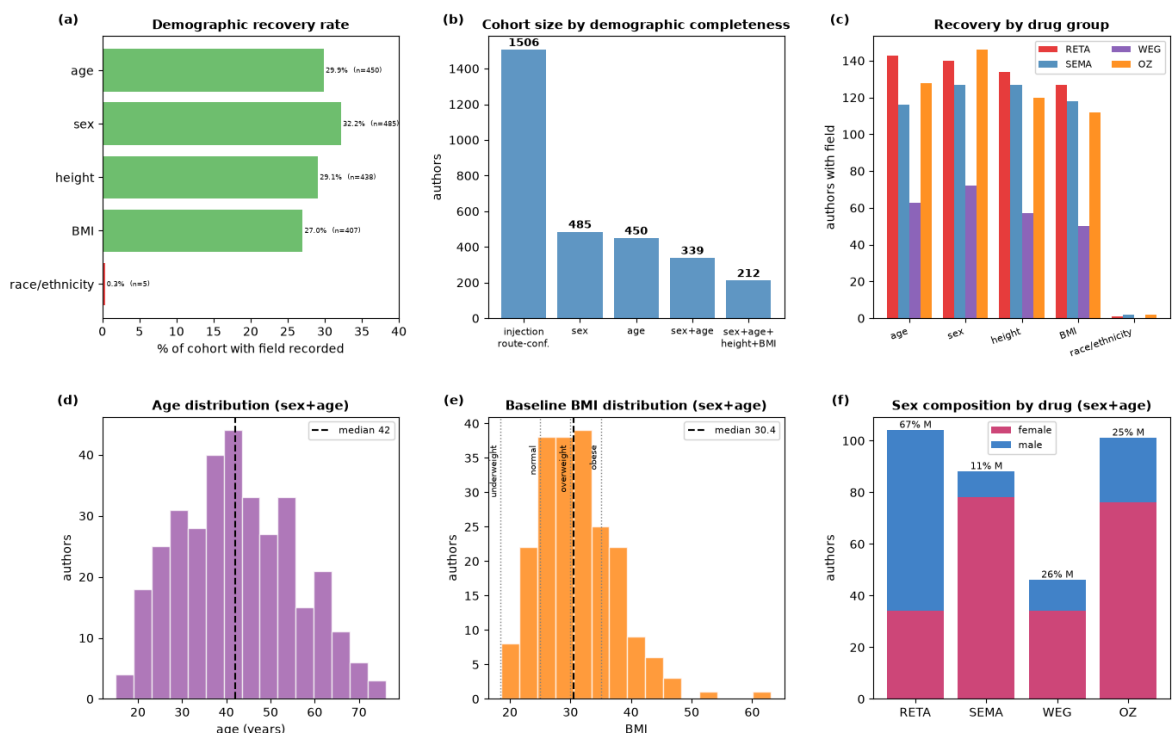
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/3952683660.py:71: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator. Otherwise, ticks may be mislabeled.

```

ax6.set_ylabel("authors"); ax6.set_xticklabels([d.upper() for d in DRUG_ORDER])

```

GLP-1 Reddit cohort — demographic recovery & composition



```

In [ ]: #####
#BUBBLE HEATMAP: reta vs sema across demographic subgroups #
#####
ADVERSE_6 = ["gastrointestinal issues", "dose intolerance", "elevated heart rate",
    "skin sensitivity", "weight gain", "muscle loss"]
DESIRED_6 = ["weight loss", "relieved gastrointestinal symptoms", "reduce food intake",
    "reduced food noise", "improved quality of life", "improved sleep"]
COLUMNS = [(e, "adverse") for e in ADVERSE_6] + [(e, "desired") for e in DESIRED_6]

age_median = np.median([age_map[a] for a in has_both if age_map.get(a) is not None])
bmi_median = np.median([bmi_map[a] for a in has_both if bmi_map.get(a) is not None])
ROWS = ["All", "Males", "Females",
    f"Age<{age_median:.0f}", f"Age≥{age_median:.0f}",

```

```

        f"BMI<{bmi_median:.1f}", f"BMI≥{bmi_median:.1f}"]

def subgroup_members(name):
    """sex+age author-ids belonging to a named demographic subgroup row."""
    if name == "All": return list(has_both)
    if name == "Males": return [a for a in has_both if sex_map.get(a) == "M"]
    if name == "Females": return [a for a in has_both if sex_map.get(a) == "F"]
    if name.startswith("Age<"): return [a for a in has_both if age_map.get(a) < age_median]
    if name.startswith("Age≥"): return [a for a in has_both if age_map.get(a) ≥ age_median]
    if name.startswith("BMI<"): return [a for a in has_both if bmi_map.get(a) < bmi_median]
    if name.startswith("BMI≥"): return [a for a in has_both if bmi_map.get(a) ≥ bmi_median]
    return []

def cell_stats(effect, event, members):
    """Per subgroup cell: reta-sema rate difference, pooled rate, Fisher's test p-value"""
    reta = [a for a in members if _drug.get(a) == "reta"]
    sema = [a for a in members if _drug.get(a) == "sema"]
    if not reta or not sema:
        return None
    reta_k = authors_reporting(effect, event, reta); reta_n = len(reta)
    sema_k = authors_reporting(effect, event, sema); sema_n = len(sema)
    _, p = fisher_exact([reta_k, reta_n-reta_k], [sema_k, sema_n-sema_k])
    return dict(diff=reta_k/reta_n - sema_k/sema_n, pooled=(reta_k+sema_k)/(reta_n+sema_n), p=p)

grid = {}; pvals_flat = []
for r, row in enumerate(ROWS):
    members = subgroup_members(row)
    for c, (effect, event) in enumerate(COLUMNS):
        st = cell_stats(effect, event, members)
        grid[(r, c)] = st
        if st is not None:
            pvals_flat.append((r, c, st["p"]))
qmap = dict(zip([k for k, _ in pvals_flat],
                multipletests([p for _, p in pvals_flat], method="fdr_bh")))

fig, ax = plt.subplots(figsize=(15, 9))
norm = TwoSlopeNorm(vmin=-0.4, vcenter=0, vmax=0.4); cmap = plt.cm.RdBu_r
for (r, c), st in grid.items():
    if st is None:
        ax.add_patch(plt.Rectangle((c-0.5, r-0.5), 1, 1, color="#e0e0e0"))
        ax.scatter(c, r, s=200+2600*st["pooled"], color=cmap(norm(st["diff"])),
                    edgecolors="black", linewidths=0.7, zorder=2)
    if qmap.get((r, c), 1) < 0.05:
        ax.text(c, r, "*", ha="center", va="center", fontsize=13, fontweight="bold")
ax.set_yticks(range(len(ROWS))); ax.set_yticklabels(ROWS, fontsize=10); ax.set_xticks(range(len(COLUMNS)))
ax.set_xticklabels([e for e, _ in COLUMNS], rotation=40, ha="left", rotate=True)
ax.xaxis.tick_top(); ax.xaxis.set_label_position("top"); ax.tick_params(axis="x", labeltop=True)
ax.set_xlim(-0.6, len(COLUMNS)-0.4)
ax.axvline(len(ADVERSE_6)-0.5, color="black", lw=1.5) # adverse
ax.text((len(ADVERSE_6)-1)/2, 1.3, "ADVERSE", ha="center", va="bottom", fontsize=13, transform=ax.get_xaxis_transform())
ax.text(len(ADVERSE_6)+(len(DESIRED_6)-1)/2, 1.3, "DESIRED", ha="center", va="bottom", fontsize=13, transform=ax.get_xaxis_transform())
sm = ScalarMappable(norm=norm, cmap=cmap); sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, fraction=0.03, pad=0.02)
cbar.set_label("reta - sema difference in reporting rate\n(red = reta high rate, blue = sema high rate)")
size_handles = [plt.scatter([], [], s=200+2600*rate, color="lightgrey", edgecolors="black", linewidths=0.8, label=f"{int(rate*100)}%") for rate in [0.0, 0.2, 0.4, 0.6, 0.8, 1.0]]
size_legend = ax.legend(handles=size_handles, title="pooled reporting rate", loc="upper right", border=False)

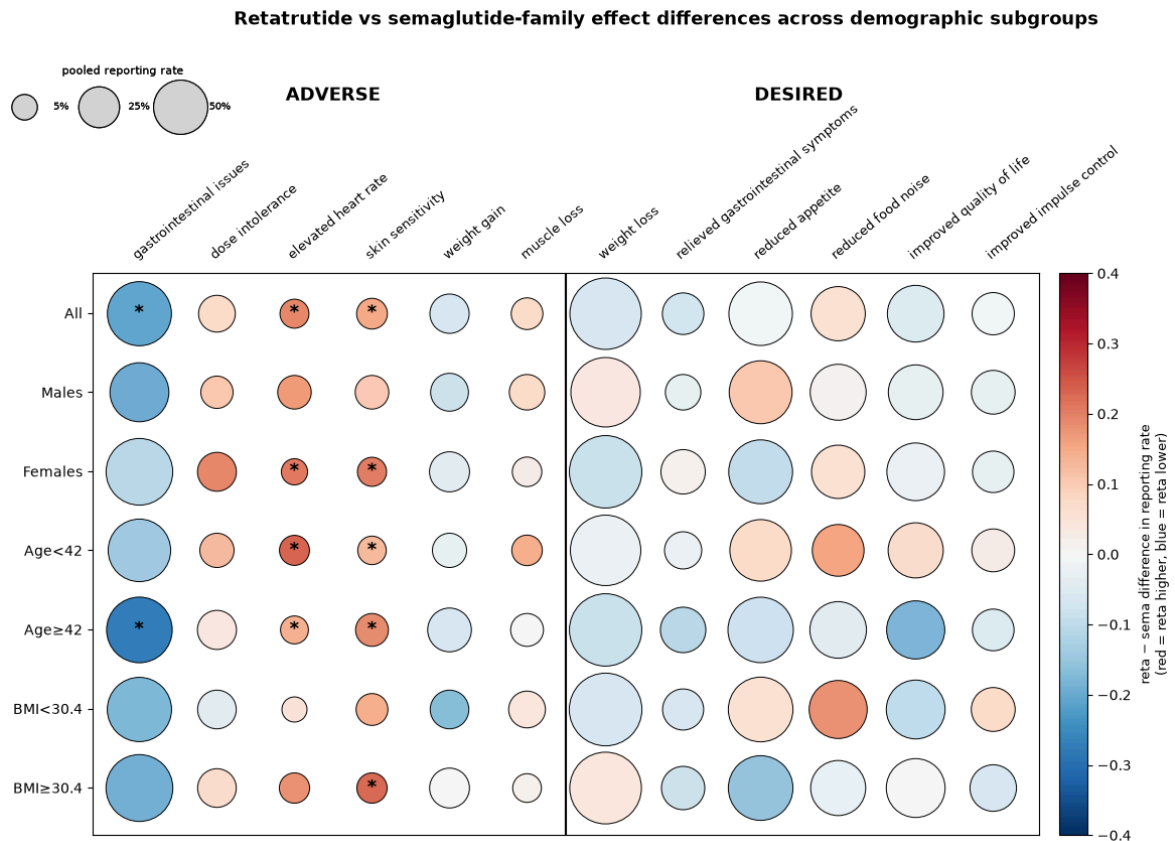
```



```

loc="upper left", bbox_to_anchor=(-0.1, 1.4),
fontsize=7, title_fontsize=8, labelspace=2.5,
borderpad=1.2, handletextpad=2, frameon=False,
ncol=3)
ax.add_artist(size_legend); ax.grid(False)
plt.subplots_adjust(top=0.68, left=0.12, right=0.78, bottom=0.06)
fig.suptitle("Retatrutide vs semaglutide-family effect differences across
             fontsize=13, fontweight="bold", y=0.97)
plt.savefig(f"{gpt_result_dir}/bubble_heatmap_subgroups.png", dpi=140)
plt.show()

```



```

In [ ]: # #####
# PED vs nonPED AE+DE HEATMAP #
#####

subgroups = {"reta-PED": reta_ped, "reta-nonPED": reta_noped,
             "sema-PED": sema_ped, "sema-nonPED": sema_noped}
subgroup_sizes = {name: len(ids) for name, ids in subgroups.items()}

def build_panel(effects, event):
    """Prevalence grid (effects x 4 subgroups) + BH q of the deconfounded
    table = pd.DataFrame({name: [prevalence(e, event, ids) for e in effects
                                for name, ids in subgroups.items()}], index=effects)
    pvals = []
    for eff in effects:
        reta_k = authors_reporting(eff, event, reta_noped)
        sema_k = authors_reporting(eff, event, sema_noped)
        _, p = fisher_exact([[reta_k, len(reta_noped)-reta_k], [sema_k, len(sema_noped)-sema_k]])
        pvals.append(p)
    return table, multipletests(pvals, method="fdr_bh")[1]

adv_table, adv_q = build_panel(top_reta_effects("adverse", 11), "adverse")
des_table, des_q = build_panel(top_reta_effects("desired", 11), "desired")

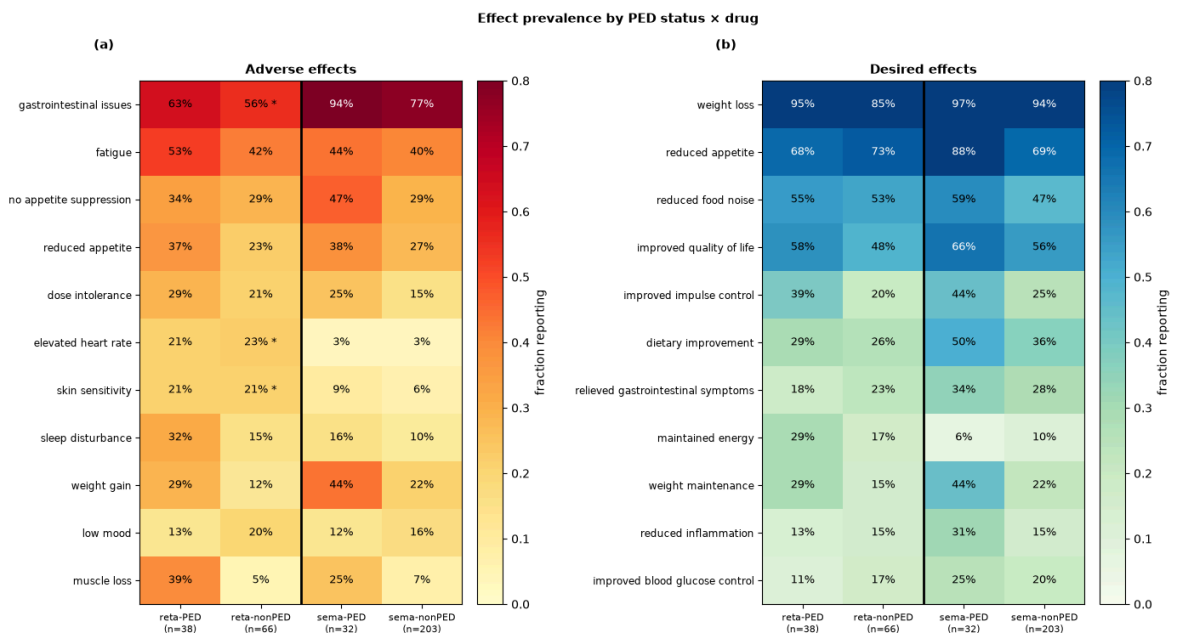
```

```

def draw_panel(ax, table, qvals, cmap, title, vmax=0.8):
    im = ax.imshow(table.values, aspect="auto", cmap=cmap, vmin=0, vmax=v
    ax.axvline(1.5, color="black", lw=2) # reta
    ax.set_xticks(range(len(subgroups)))
    ax.set_xticklabels([f"{n}\n(n={subgroup_sizes[n]})" for n in subgroup
    ax.set_yticks(range(len(table.index))); ax.set_yticklabels(table.inde
    reta_col = list(subgroups).index("reta-nonPED")
    for row in range(len(table.index)):
        for col in range(len(subgroups)):
            value = table.values[row, col]
            if np.isnan(value): continue
            star = " *" if (col == reta_col and qvals[row] < 0.05) else ""
            ax.text(col, row, f"{value*100:.0f}%{star}", ha="center", va=
                    fontsize=9, color="black" if value < vmax*0.8 else "w
    ax.set_title(title, fontsize=11, fontweight="bold")
    return im

fig, (ax_adv, ax_des) = plt.subplots(1, 2, figsize=(15, 8))
plt.subplots_adjust(wspace=0.7)
im_adv = draw_panel(ax_adv, adv_table, adv_q, "YlOrRd", "Adverse effects"
im_des = draw_panel(ax_des, des_table, des_q, "GnBu", "Desired effects"
fig.colorbar(im_adv, ax=ax_adv, fraction=0.08, pad=0.04, label="fraction
fig.colorbar(im_des, ax=ax_des, fraction=0.08, pad=0.04, label="fraction
for label, ax in zip("ab", [ax_adv, ax_des]):
    ax.text(-0.08, 1.08, f"({label})", transform=ax.transAxes,
            fontsize=11, fontweight="bold", va="top", ha="right")
fig.suptitle("Effect prevalence by PED status × drug", fontsize=11, fontw
plt.savefig(f"{gpt_result_dir}/ped_deconfound_heatmap_top11.png", dpi=140
plt.show()

```



```

In [ ]: #####
# EVENT FREQUENCY: PED vs non-PED (sema pooled + by drug) #
#####
def author_event_counts(aid):
    """Per-author reported effect-event counts: total, adverse, desired."""
    p = _by.get(aid) or {}; adverse = desired = 0
    for e in (p.get("timeline_events") or []):
        if isinstance(e, dict):
            adverse += len([x for x in (e.get("adverse_effects") or []) i
            desired += len([x for x in (e.get("desired_effects") or []) i

```

```

    return {"total": adverse + desired, "adverse": adverse, "desired": de

freq = pd.DataFrame([{"author_id": a, "drug": _drug.get(a), "ped": uses_p
                      for a in has_both])

def med_iqr(values):
    """'median (IQR lo-hi)' string for a set of counts."""
    values = np.asarray([v for v in values if v is not None and not np.is
    return "-" if len(values) == 0 else f"{np.median(values):.1f} ({np.pe

def mwu_p(a, b):
    """Mann-Whitney U two-sided p (skewed counts); nan if either group em
    a = [v for v in a if v is not None and not np.isnan(v)]; b = [v for v
    return np.nan if not a or not b else mannwhitneyu(a, b, alternative="

def comparison_table(subset, label):
    """PED vs non-PED median(IQR) + MWU p for total/adverse/desired count
    ped = subset[subset["ped"]]; noped = subset[~subset["ped"]]
    rows = [{"metric": m, "PED med (IQR)": med_iqr(ped[m]), "nonPED med (
              "MWU p": mwu_p(ped[m].values, noped[m].values)} for m in ["t
    df = pd.DataFrame(rows)
    df["MWU p"] = df["MWU p"].map(lambda v: f"{v:.3f}" if not np.isnan(v)
    df["sig"]    = [("*" if p != "-" and float(p) < 0.05 else "") for p in
    df.insert(0, "cohort", f"{label} (PED={len(ped)}, nonPED={len(noped)})
    return df

tables = [comparison_table(freq, "POOLED (all drugs)")]
for drug in DRUG_ORDER:
    sub = freq[freq["drug"] == drug]
    if sub["ped"].any() and (~sub["ped"]).any():
        tables.append(comparison_table(sub, drug))
print("Event frequency per author – PED vs non-PED (sex+age cohort)\n")
print(pd.concat(tables, ignore_index=True).to_string(index=False))

```

Event frequency per author – PED vs non-PED (sex+age cohort)

	cohort	metric	PED med (IQR)	nonPED me
d (IQR) MWU p sig				
P00LED (all drugs) (PED=70, nonPED=269)	total	29.5 (14.2–74.8)	20.0 (10.0–37.0)	0.001 *
P00LED (all drugs) (PED=70, nonPED=269)	adverse	15.0 (4.2–29.5)	9.0 (4.0–17.0)	0.005 *
P00LED (all drugs) (PED=70, nonPED=269)	desired	16.5 (9.0–35.2)	11.0 (5.0–22.0)	0.002 *
reta (PED=38, nonPED=66)	total	23.0 (10.5–47.0)	13.5 (8.0–28.0)	0.033 *
reta (PED=38, nonPED=66)	adverse	8.5 (3.0–25.2)	6.0 (2.0–14.0)	0.072
reta (PED=38, nonPED=66)	desired	12.5 (5.0–27.2)	7.0 (4.0–15.5)	0.022 *
sema (PED=15, nonPED=73)	total	46.0 (24.5–81.0)	23.0 (14.0–44.0)	0.031 *
sema (PED=15, nonPED=73)	adverse	18.0 (7.5–31.5)	10.0 (5.0–19.0)	0.075
sema (PED=15, nonPED=73)	desired	22.0 (13.5–39.0)	13.0 (6.0–27.0)	0.149
weg (PED=2, nonPED=44)	total	61.5 (52.8–70.2)	14.5 (8.0–27.0)	0.038 *
weg (PED=2, nonPED=44)	adverse	27.5 (24.2–30.8)	6.0 (2.8–12.8)	0.049 *
weg (PED=2, nonPED=44)	desired	34.0 (28.5–39.5)	8.5 (4.0–14.0)	0.043 *
oz (PED=15, nonPED=86)	total	48.0 (20.5–91.5)	25.0 (15.0–45.2)	0.059
oz (PED=15, nonPED=86)	adverse	24.0 (6.5–36.0)	11.0 (5.0–18.8)	0.061
oz (PED=15, nonPED=86)	desired	19.0 (11.5–52.0)	13.5 (6.2–26.5)	0.076

```
In [ ]: #####
# SWIMMER PLOTS (for report: 6 AEs + 6 DEs) #
# Author-level temporal reconstruction.      #
#####

# vertical offset: adverse markers above the lane, desired below
OFFSET = 0.18

STYLE_6 = {
    # adverse (above the lane) – reds
    ("gastrointestinal issues", "adverse"): ("^", "#b2182b"),
    ("dose intolerance", "adverse"):         ("v", "#d6604d"),
    ("elevated heart rate", "adverse"):       ("X", "#d73027"),
    ("skin sensitivity", "adverse"):          ("P", "#fc8d59"),
    ("weight gain", "adverse"):               ("s", "#8c510a"),
    ("muscle loss", "adverse"):               ("h", "#7f0000"),
    # desired (below the lane) – greens
    ("weight loss", "desired"):               ("o", "#1a9850"),
    ("relieved gastrointestinal symptoms", "desired"): ("^", "#00806a"),
    ("reduced appetite", "desired"):          ("D", "#4daf4a"),
    ("reduced food noise", "desired"):        ("s", "#006837"),
    ("improved quality of life", "desired"):   ("h", "#66c2a4"),
    ("improved impulse control", "desired"):   ("P", "#35978f"),
}
```

```

def author_journey(aid, hl_adverse, hl_desired):
    """Reconstruct one author's journey, keeping only the highlighted eff
    For each timeline event with a date, calculate the number of weeks si
    author's FIRST dated event (week = days_since_first / 7), then count
    for every adverse or desired effect from CANONICAL labels in the high
    set.

    Returns a dict:
        "duration"      : weeks from the first to the last dated event (jou
        "hits"          : list of (week, canonical_effect, event_type) for
        "adverse_count" : number of highlighted ADVERSE hits (used by the
        "desired_count" : number of highlighted DESIRED hits (used by the
    Returns None if the author has no dated events."""
    p = _by.get(aid) or {}
    events = [e for e in (p.get("timeline_events") or []) if isinstance(e
    if not events:
        return None
    dates = pd.to_datetime([e["date"] for e in events], errors="coerce")
    first = dates.min()
    if pd.isna(first):
        return None
    hits = []; adverse_count = desired_count = 0; weeks_all = []
    for e, d in zip(events, dates):
        if pd.isna(d):
            continue
        week = (d - first).days / 7.0
        weeks_all.append(week)
        for it in (e.get("adverse_effects") or []):
            if isinstance(it, dict):
                name = canonical_label(it.get("visualisation_group") or "
                if name in hl_adverse:
                    hits.append((week, name, "adverse")); adverse_count +
        for it in (e.get("desired_effects") or []):
            if isinstance(it, dict):
                name = canonical_label(it.get("visualisation_group") or "
                if name in hl_desired:
                    hits.append((week, name, "desired")); desired_count +
    if not weeks_all:
        return None
    return {"duration": max(weeks_all), "hits": hits,
            "adverse_count": adverse_count, "desired_count": desired_count}

def draw_swimmer(drug, style, hl_adverse, hl_desired, per_drug, min_wk, m
                min_des=0, title_extra="", fname=""):
    """One swimmer plot per drug. Lane bar black=PED / drug-colour=non-PE
    Selects the longest qualifying journeys within min_wk, max_wk range w
    least min_des highlighted desired events, up to per_drug authors."""
    journeys = []
    for a in [x for x in has_both if _drug.get(x) == drug]:
        j = author_journey(a, hl_adverse, hl_desired)
        if j and min_wk <= j["duration"] <= max_wk and j["desired_count"]
            journeys.append((a, j))
    journeys.sort(key=lambda t: t[1]["duration"], reverse=True)
    journeys = journeys[:per_drug]
    if not journeys:
        print(f"{DRUG_NAME[drug]}: no qualifying authors"); return
    adv_handles = [Line2D([0],[0], marker=m, color="w", markerfacecolor=
                        linestyle="None", markersize=9, label=e)
                    for (e, ev), (mk, c) in style.items() if ev == "advers

```

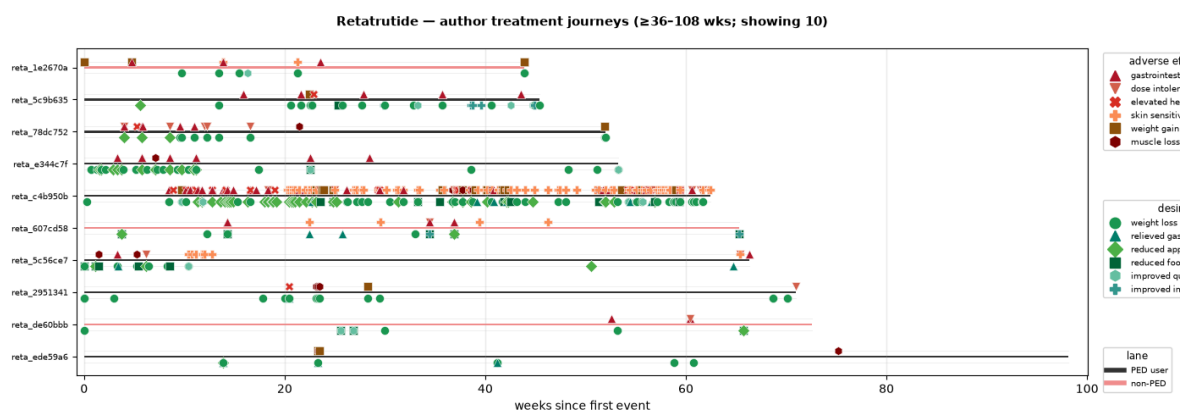
```

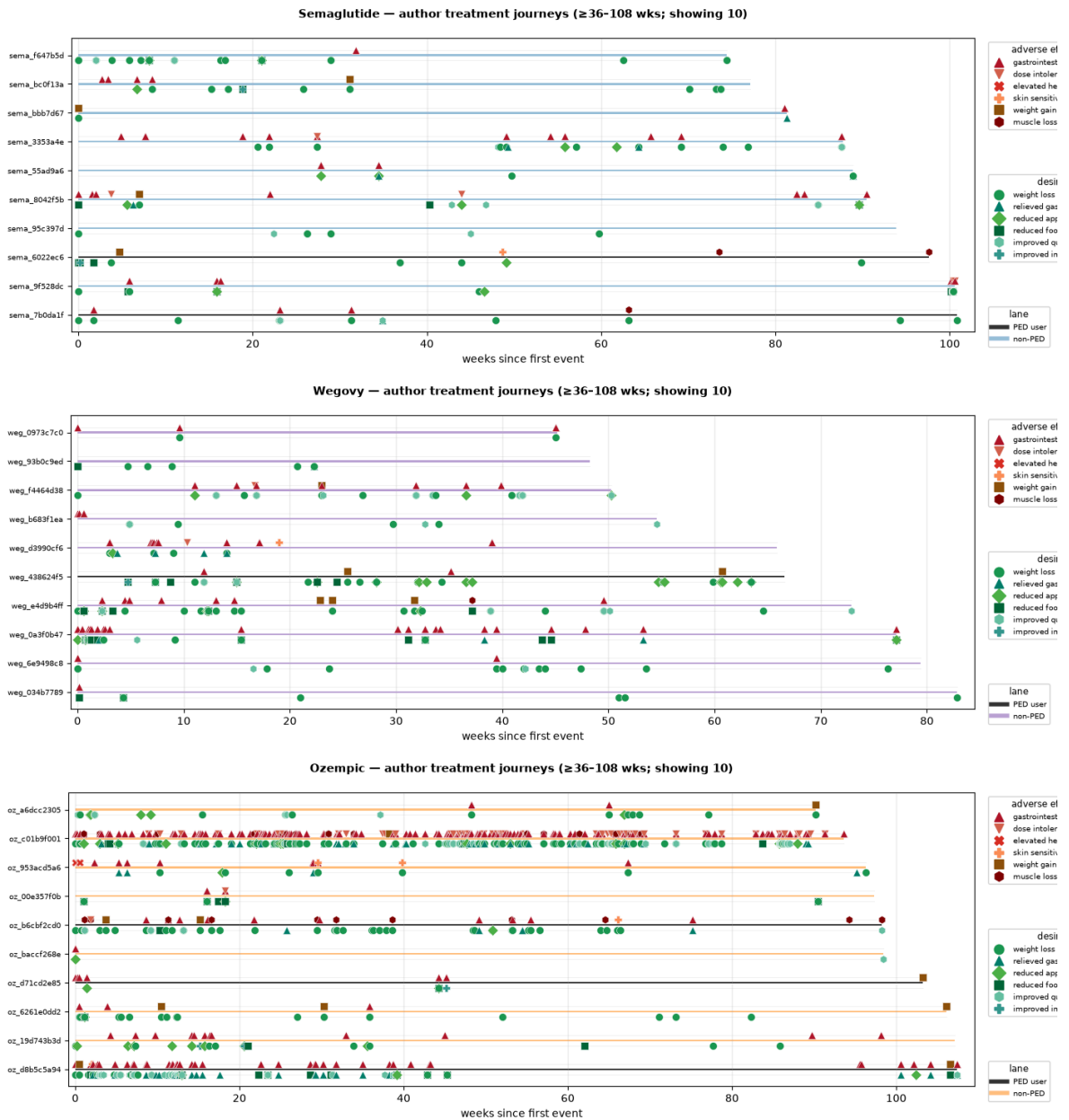
des_handles = [Line2D([0],[0], marker=mk, color="w", markerfacecolor=
                    linestyle="None", markersize=9, label=e)
                for (e, ev), (mk, c) in style.items() if ev == "desire

fig, ax = plt.subplots(figsize=(14, max(4, 0.5*len(journeys))))
for lane, (aid, j) in enumerate(journeys):
    is_ped = uses_ped(aid)
    lane_colour = "black" if is_ped else DRUG_COLOUR[drug]
    ax.barh(lane, j["duration"], height=0.06, color=lane_colour,
            alpha=0.8 if is_ped else 0.5, zorder=1)
    ax.plot([0, j["duration"]], [lane+OFFSET]*2, color="#eeeeee", lw=
    ax.plot([0, j["duration"]], [lane-OFFSET]*2, color="#eeeeee", lw=
    for week, effect, event in j["hits"]:
        marker, colour = style[(effect, event)]
        y = lane + OFFSET if event == "adverse" else lane - OFFSET
        ax.scatter(week, y, marker=marker, color=colour, s=52, zorder
                    edgecolors="white", linewidths=0.4, clip_on=False)
    ax.set_yticks(range(len(journeys))); ax.set_yticklabels([a[:12] for a
    max_dur = max(j["duration"] for _, j in journeys)
    ax.set_xlim(-max_dur*0.008, max_dur*1.02) # small left margin so we
    ax.set_ylim(-0.6, len(journeys)-0.4)
    ax.set_xlabel("weeks since first event")
    ax.set_title(f"{DRUG_NAME[drug]} – author treatment journeys ({title_
                    fontsize=11, fontweight="bold", color="black")
    ax.grid(axis="x", alpha=0.2)
    lane_handles = [Line2D([0],[0], color="black", lw=4, alpha=0.8, label
                    Line2D([0],[0], color=DRUG_COLOUR[drug], lw=4, alpha=
    leg_adv = ax.legend(handles=adv_handles, title="adverse effect", loc=
                    bbox_to_anchor=(1.01, 1.0), fontsize=7.5, title_f
    leg_des = ax.legend(handles=des_handles, title="desired effect", loc=
                    bbox_to_anchor=(1.01, 0.55), fontsize=7.5, title_
    ax.legend(handles=lane_handles, title="lane", loc="upper left",
                    bbox_to_anchor=(1.01, 0.10), fontsize=7.5, title_fontsize=8
    plt.tight_layout()
    plt.savefig(f"{gpt_result_dir}/{fname}_{drug}.png", dpi=140, bbox_inc
    plt.show()

for drug in DRUG_ORDER:
    draw_swimmer(drug, STYLE_6, set(ADVERSE_6), set(DESIRED_6), per_drug=
                    min_wk=36, max_wk=108, title_extra="≥36–108 wks", fname=

```





```
In [ ]: #####
# Stacked swimmer #
#####

# panel order: (a) reta, (b) sema, (c) weg, (d) oz
panel_drugs = ["reta", "sema", "weg", "oz"]
panel_labels = ["a", "b", "c", "d"]
panel_paths = [f"{gpt_result_dir}/swimmer_main_{dg}.png" for dg in panel_drugs]

# load images and size the stacked canvas to their combined aspect ratio
images = [mpimg.imread(p) for p in panel_paths]
# each image is roughly the same width; total height = sum of heights
heights = [im.shape[0] for im in images]
widths = [im.shape[1] for im in images]
fig_w = 15
fig_h = fig_w * (sum(heights) / max(widths))

fig, axes = plt.subplots(4, 1, figsize=(fig_w, fig_h))
for ax, im, label in zip(axes, images, panel_labels):
    ax.imshow(im)
    ax.axis("off")
    # panel label in the top-left corner of each image
```



```

        ax.text(0.01, 0.97, f"({label})", transform=ax.transAxes,
               fontsize=8, fontweight="bold", va="top", ha="left")

plt.subplots_adjust(hspace=0.03, left=0.005, right=0.995, top=0.995, bottom=0.005)
plt.savefig(f"{gpt_result_dir}/swimmer_stacked_abcd.png", dpi=150, bbox_inches='tight')
plt.show()
print("saved stacked figure -> swimmer_stacked_abcd.png")

```

```

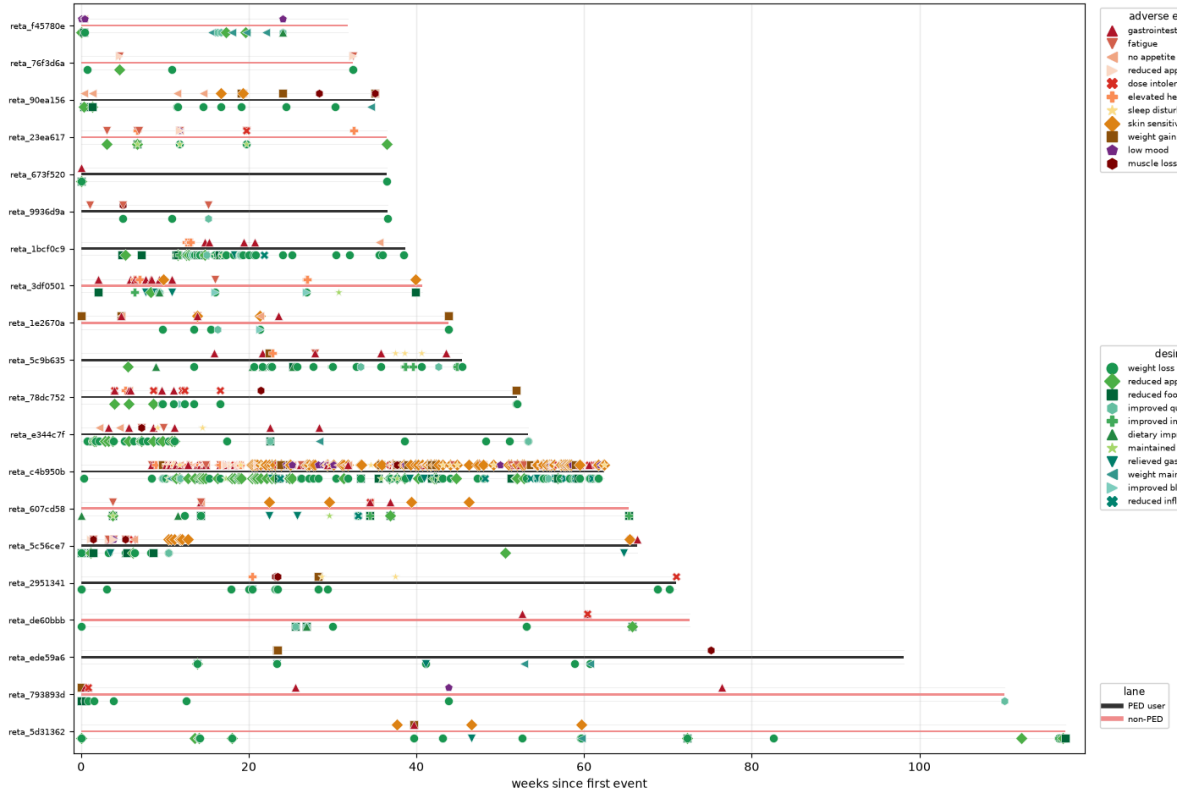
In [ ]: #####
# EXTENDED SWIMMER PLOTS (11 AEs + 11 DEs) #
#####

HL_ADVERSE_11 = set(top_reta_effects("adverse", 11))
HL_DESIRE_11 = set(top_reta_effects("desired", 11))
STYLE_11 = {
    ("gastrointestinal issues", "adverse"): ("^", "#b2182b"),
    ("fatigue", "adverse"): ("v", "#d6604d"),
    ("no appetite suppression", "adverse"): ("<", "#f4a582"),
    ("reduced appetite", "adverse"): (">", "#fddbc7"),
    ("dose intolerance", "adverse"): ("X", "#d73027"),
    ("elevated heart rate", "adverse"): ("P", "#fc8d59"),
    ("sleep disturbance", "adverse"): ("*", "#fee08b"),
    ("skin sensitivity", "adverse"): ("D", "#e08214"),
    ("weight gain", "adverse"): ("s", "#8c510a"),
    ("low mood", "adverse"): ("p", "#762a83"),
    ("muscle loss", "adverse"): ("h", "#f00000"),
    ("weight loss", "desired"): ("o", "#1a9850"),
    ("reduced appetite", "desired"): ("D", "#4daf4a"),
    ("reduced food noise", "desired"): ("s", "#006837"),
    ("improved quality of life", "desired"): ("h", "#66c2a4"),
    ("improved impulse control", "desired"): ("P", "#41ab5d"),
    ("dietary improvement", "desired"): ("^", "#238b45"),
    ("maintained energy", "desired"): ("*", "#a6d96a"),
    ("relieved gastrointestinal symptoms", "desired"): ("v", "#00806a"),
    ("weight maintenance", "desired"): ("<", "#35978f"),
    ("improved blood glucose control", "desired"): (">", "#80cdc1"),
    ("reduced inflammation", "desired"): ("X", "#018571"),
}

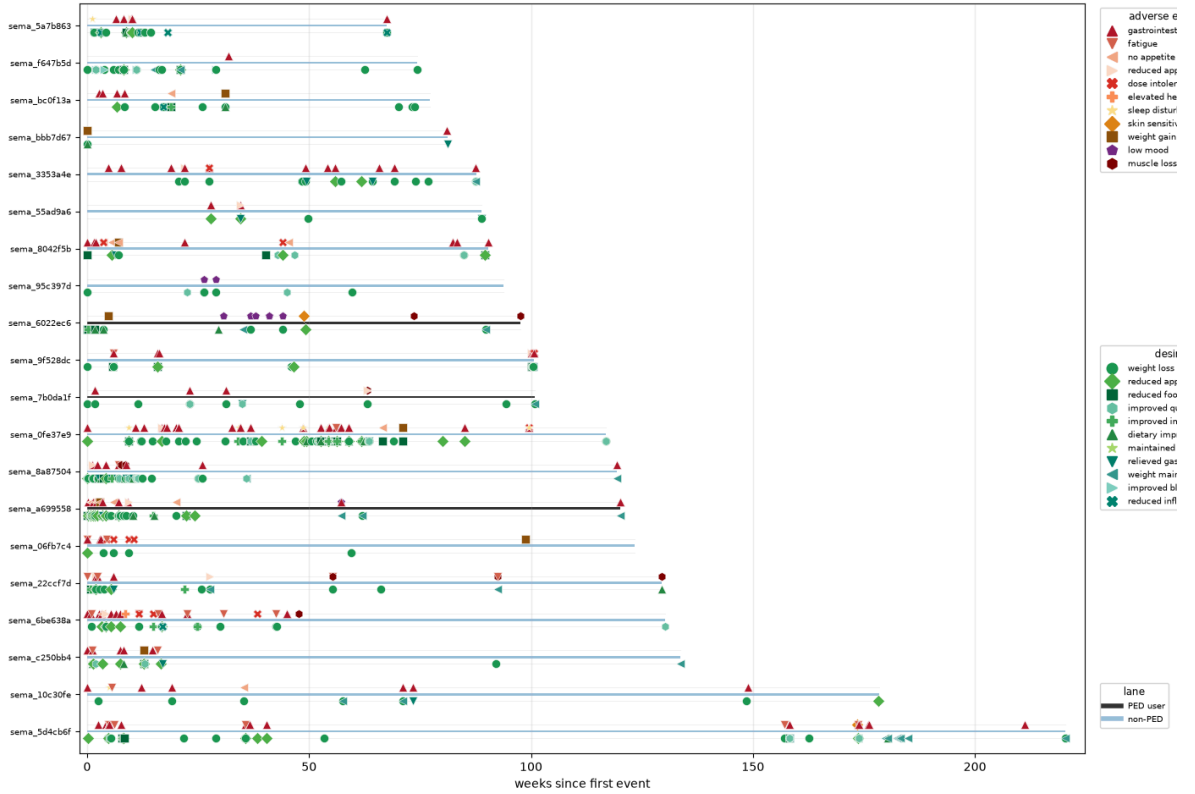
for drug in DRUG_ORDER:
    draw_swimmer(drug, STYLE_11, HL_ADVERSE_11, HL_DESIRE_11, per_drug=2,
                  min_wk=12, max_wk=10_000, min_des=3,
                  title_extra="sex+age, ≥12 wks & ≥3 desired", fname="swim

```

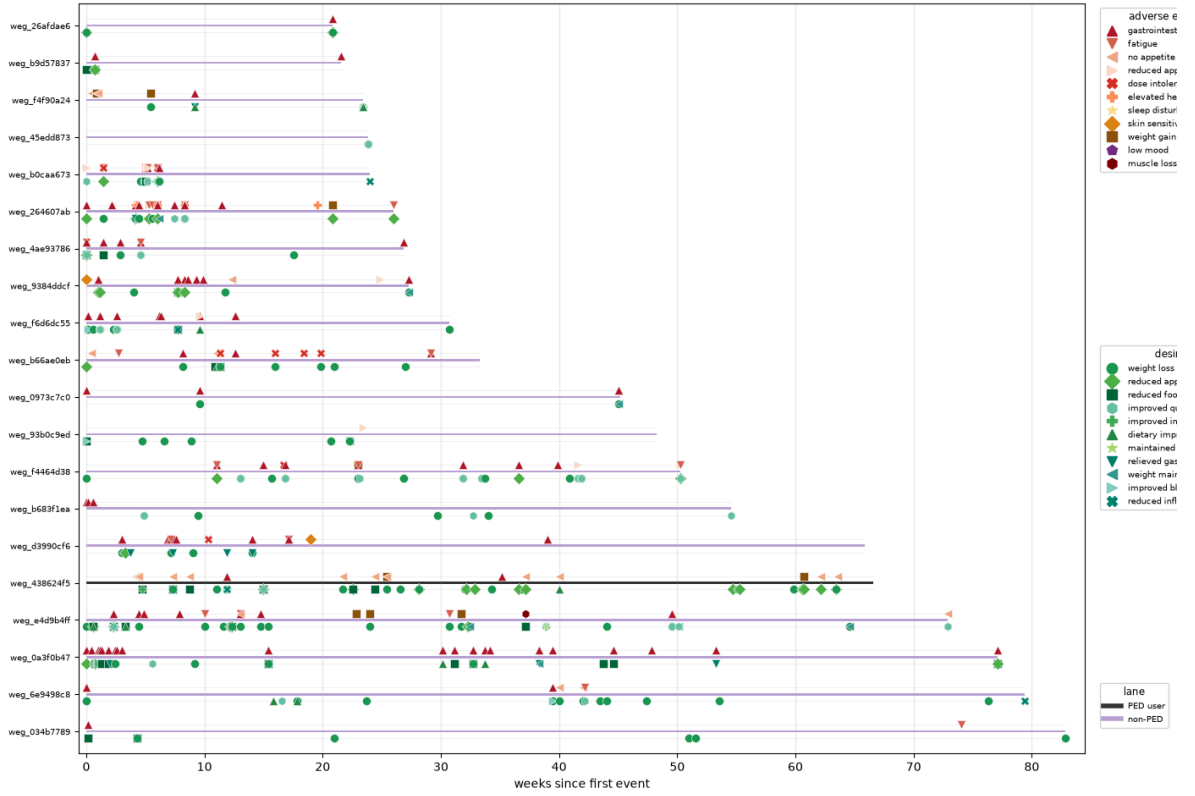

Retatrutide — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)



Semaglutide — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)



Wegovy — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)



Ozempic — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)

